

Thesis

Vehicle-in-the-Loop simulation using CARLA simulator

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Abstract

This thesis investigates the design, implementation, and evaluation of a Lane Keeping Assist (LKA) function inside a software-only Vehicle-in-the-Loop simulation environment based on the CARLA simulator. The work develops a complete control stack that runs as an external Python process, communicates with the simulator through its synchronous-mode Python API at 20 Hz, and exercises the same vehicle, the same map, and the same target speed across three progressively richer lateral-control architectures: a cross-track PID closed on the perpendicular distance to a pre-recorded reference path, a heading-error PI closed on the angle to a speed-dependent look-ahead point on the same reference, and a vision-based PI that replaces the pre-recorded reference with a centre line reconstructed at every tick from the RGB camera by a deep convolutional lane detector and by an inverse-perspective-mapping projection.

The vehicle is modelled as a dynamic single-track with parameters read from the CARLA blueprint, and a complete system-identification procedure fits the longitudinal time constant and the lateral steering gain from two open-loop step experiments performed in the simulator. The resulting plant parameters are used to derive analytical PID gains by pole placement, with explicit gain-scheduling expressions that show that the cross-track gains scale as $1/v^2$ while the heading-error gains scale as $1/v$ —a structural advantage of the heading-error formulation that is reflected in a larger closed-loop phase margin and in a gentler dependence on the vehicle speed.

The three architectures are evaluated on a common test route in Town06 with two scalar metrics: the root-mean-square cross-track error (RMSE) and the maximum absolute cross-track error. The heading-error PI achieves the best performance with an RMSE of 0.120 m and a maximum excursion of 1.185 m, against 0.205 m and 2.217 m of the cross-track PID—an improvement by a factor of approximately 1.7 on the average and 1.9 on the worst case, consistent with the control-theoretic predictions on the reduced plant order. The vision-based PI achieves an RMSE of 0.270 m, dominated by a steady-state bias of approximately 0.7 m on smooth curves that is traced back to specific approximations of the perception layer. On a step-like lane discontinuity, the vision-based controller is shown to produce the smoothest physical trajectory at the cost of a large *apparent* error against the discontinuous path-based reference.

The thesis contributes a modular, reproducible software-only Vehicle-in-the-Loop implementation of an LKA function, a complete system-identification and analytical tuning pipeline, a side-by-side control-theoretic and experimental comparison of three lateral-control architectures, and an explicit junction-handling logic that makes the vision-based controller usable on routes that include intersections. The closing chapter outlines a structured set of short-, medium-, and long-term extensions, ranging from a curvature-aware look-ahead correction and a model-predictive controller, through sensor fusion between camera and map, to real-vehicle validation and learning-based control policies.

Chapter 1

Introduction

1.1 What is autonomous driving

For 3 million years human being has been creating tools to ease their daily chores in order to make better life. One of the successful creation of human as decent invention is vehicles. First automobile ever created is involved to late 19th century. Since that vehicles have been evolving rapidly, so, recent decades we had new concept of vehicle so-called “Autonomous Vehicle”.

Autonomous Vehicles (AVs), or self-driving cars, are vehicles as discussed earlier that can ease human daily driving tasks by taking them one by one to do it themselves.

Autonomous vehicles represent the next major step in the evolution of road transportation, extending earlier advances in vehicle automation and driver assistance systems. As sensing, computing power, and control technologies have matured, vehicles have gained the ability to monitor their surroundings, interpret complex traffic situations, and support or even replace human driving tasks. Early prototypes in the 1980s and 1990s, enabled by progress in computer vision, robotics, and embedded systems, demonstrated that vehicles could follow lanes, maintain speed, and avoid obstacles under constrained conditions. The parallel development of GPS-based positioning, digital mapping, and onboard sensors such as cameras, radars, and lidars provided the foundation for modern perception systems. Last 7 decades, Autonomous driving has got big advances which can be seen in Figure 1.1. These technological advances collectively enabled the transition from basic driver assistance toward higher levels of automation, where vehicles can execute steering, acceleration, and braking tasks with increasing independence from human input. Driverless vehicle technology gained momentum at the beginning of the 21st century. Initiatives like Google’s driverless car project [1] captured public attention and created a general awareness that autonomous driving was a real possibility. The use of cameras, radars, lidars, and artificial intelligence algorithms in vehicles endowed them with the capability to navigate complex traffic situations independently. Many car manufacturers and technology companies are now accelerating their efforts to develop fully autonomous vehicles. The goal of these vehicles is to reduce traffic accidents, increase transportation efficiency, and provide greater independence for people with mobility limitations. However, technical challenges, as well as ethical, legal, and safety issues, continue to shape the advancement in this field. The future of autonomous driving depends on overcoming these challenges and gaining societal acceptance for this new technology.

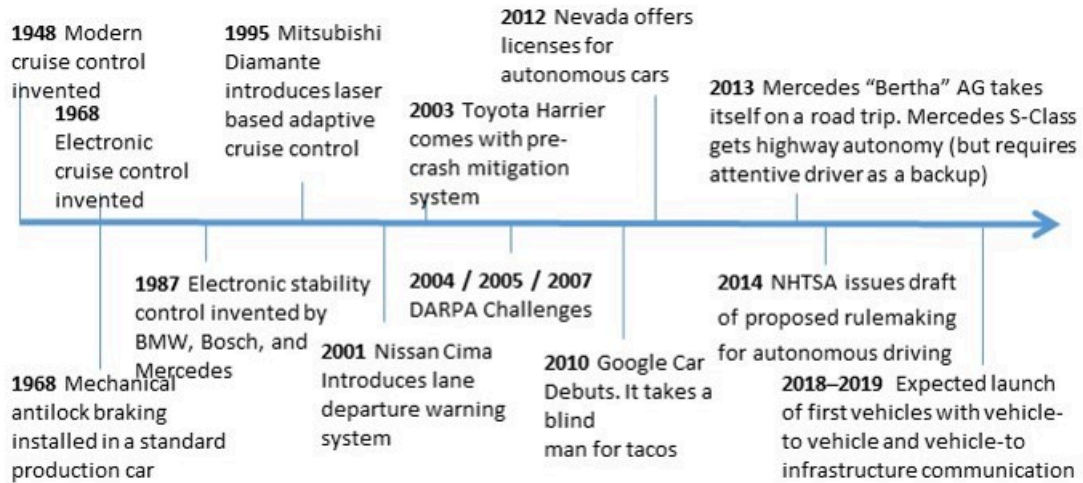


Figure 1.1: The historical development of autonomous driving [2].

1.2 Importance of Simulation in Autonomous Driving

In today's world, developing and using software has become very important across many industries. While these software solutions provide us with numerous conveniences and advantages, they also introduce certain requirements. One of the most critical requirements for any software is testing. The testing phase can be used to determine whether an application fulfills its intended function or to assess its efficiency. Of course, in some sectors, this functionality takes on an even greater importance. One such area is safety. In industries like aviation and automotive, where a user's safety can be directly affected by an error in the application, testing and simulation are more important than ever. In model-based software applications, after a model is created, it moves to the verification stage. Before being loaded onto the hardware, the model undergoes several verification steps (as V model in Figure 1.2). Some of these verifications are named Model In the Loop (MIL) [3], Software In the Loop (SIL) [4], and Hardware In the Loop (HIL) [5].

Model in the Loop (MIL) In the initial stages of system design, before real hardware or software components are implemented, the simulation and testing of system models are carried out by the Model in the Loop (MIL) testing method. This method uses mathematical models and simulations for system or component design. These models are utilized to understand how the designed system will function and behave. MIL testing is a critical tool for verifying the design and identifying potential problems at an early stage. With this method, engineers and designers can gain valuable insights into how the system will operate before actually manufacturing the hardware or fully developing the software.

Software in the Loop (SIL) The Software in the Loop (SIL) testing method is the process of testing a system or component's software without using real hardware. This method simulates environmental factors or other parts interacting with the system while the software runs directly on a computer. This simulation evaluates the software's

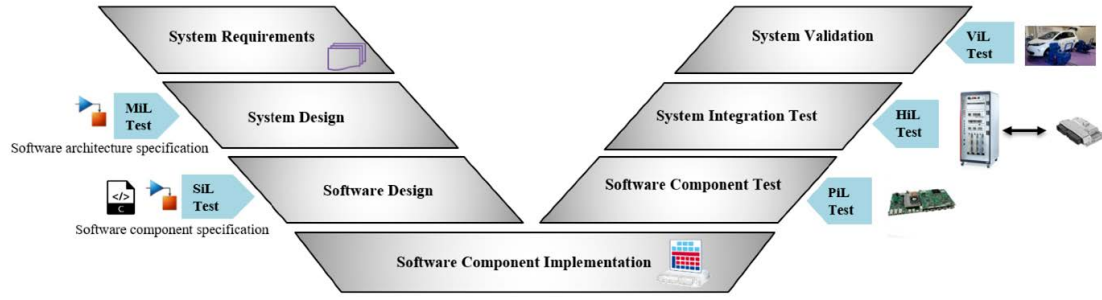


Figure 1.2: MIL, SIL , PIL, HIL and VIL tests in V-cycle development process [6].

behavior and performance under real-world conditions. The main goal of SIL testing is to verify the software’s functionality, detect errors at an early stage, and analyze interactions between the system and the software. This approach accelerates the software development and refinement processes and prevents costly errors.

Hardware in the Loop (HIL) The Hardware in the Loop (HIL) testing method creates an environment to control and test real hardware. In this method, the hardware being tested, such as a vehicle control unit, operates in real-time, but the software simulates the physical systems (e.g., engines, sensors, actuators, etc.) interacting with the hardware in real-time. This simulation allows for the evaluation of how the hardware performs under real-world conditions.

In Figure 1.3 driving test simulator is shown. Test simulators typically fall under the SIL category. The Software in the Loop (SIL) testing method has numerous advantages. These benefits explain why SIL tests are an important part of the software development process and demonstrate their value to engineers and developers across various industries. The main advantages can be listed as follows:

Cost Efficiency: SIL tests are conducted in a simulated environment that does not require real hardware, thus reducing the costs associated with purchasing, maintaining, and repairing hardware. Additionally, early diagnosis of potential errors helps prevent more costly problems later on.

Rapid Feedback Loop: Testing software in a model allows for quick iterations in the development process. This enables developers to instantly see the effects of their code and make fast modifications if necessary.

Risk Reduction: Tests conducted without real hardware use a simulated environment to safely examine situations that could be potentially dangerous or harmful to the hardware. This is especially important when expensive hardware is involved.

Broad Test Scenarios: SIL tests are capable of quickly and easily testing a wide range of scenarios that mimic real-world conditions. Developers can experiment with system parameters, error states, and various operational conditions.

Acceleration of the Development Process: SIL tests can speed up the development process instead of waiting for the hardware to be ready. Testing the software alongside hardware development shortens the time to market for the product.

Preparation for Integration and System Tests: When software is successfully tested in a SIL environment, the transition to more complex testing stages where hardware and software are run together becomes easier. This helps reduce problems that may arise during system tests and integration.



Figure 1.3: Driving Test Simulator [7]

Flexibility in the Development Process: Software can be tested and developed with various hardware platforms and configurations. This flexibility allows for the evaluation of how the software will perform on different systems.

1.3 State of the Art in Autonomous Driving Simulators

Advancements in the field of autonomous driving have also led to the development of powerful autonomous driving simulators. Nowadays, a wide range of simulators produced by various companies are in use. These include open-source projects (e.g. CARLA [8], AirSim [9]), commercial software, and customized simulation solutions used in academic and industrial research. In the context of this thesis, a simulation environment is essential, because the Lane Keeping Assist (LKA) function is developed and evaluated entirely in a virtual Vehicle-in-the-Loop setup rather than on a physical test vehicle. Therefore, it is crucial to clearly define the requirements that the simulator must satisfy—such as realistic sensor models, controllable traffic and weather, and reproducible scenarios—and to justify the choice of a specific platform.

In the following, I first review several popular autonomous driving simulation tools and platforms available in the literature and in practice. The comparison focuses on criteria such as graphic quality, accuracy of the physics engine, sensor simulation capabilities, simulation of traffic and pedestrians, weather conditions, and the ability to simulate at different times of the day. Based on these criteria, I motivate the selection of CARLA as

the main simulation platform for this work and explain how it will be used to implement and test the proposed LKA function.

1.3.1 Waymo simulator



Figure 1.4: Comparing Waymo Simulator graphics and the real world.

Waymo possesses an advanced simulation platform [10] and is widely recognized as a leader in autonomous vehicle technology. This simulator plays a crucial role in the development and testing of autonomous driving systems by providing a highly realistic (can be seen in Figure 1.4) and comprehensive virtual environment. It combines high-fidelity graphics that accurately replicate real-world environments with a precise physics engine capable of modeling vehicle dynamics, collisions, and surface interactions. Furthermore, it includes detailed sensor emulation for LIDAR, radar, and cameras, enabling a realistic representation of environmental perception. The platform can also model complex traffic flow and pedestrian behaviors, which are essential for evaluating autonomous vehicles in dense urban settings. In addition, the simulator supports a wide range of weather and lighting conditions, allowing performance assessment under rain, snow, fog, or low-light scenarios. Owing to these capabilities, the Waymo simulator holds a significant position among state-of-the-art autonomous driving simulation platforms.

1.3.2 SVL Simulator



Figure 1.5: SVL Simulator Screen

The autonomous vehicle research community frequently employs the LGSVL Simulator [11], an open-source platform designed for high-fidelity testing of perception and control algorithms. Thanks to its Unity-based rendering engine, LGSVL offers realistic visual environments suitable for evaluating camera-based functions. Its physics engine provides detailed modeling of vehicle dynamics, tire-road interactions, and collisions, enabling credible closed-loop testing of control algorithms. The simulator includes configurable sensor models—such as lidar, radar, and cameras—along with comprehensive traffic and pedestrian behavior, allowing researchers to model complex urban scenarios (Figure 1.5). Furthermore, LGSVL supports diverse weather and lighting conditions, which makes it valuable for assessing the robustness of autonomous driving functions under varying environmental circumstances. LGSVL represents an important reference point when comparing available simulation platforms.

1.3.3 Sim4CV

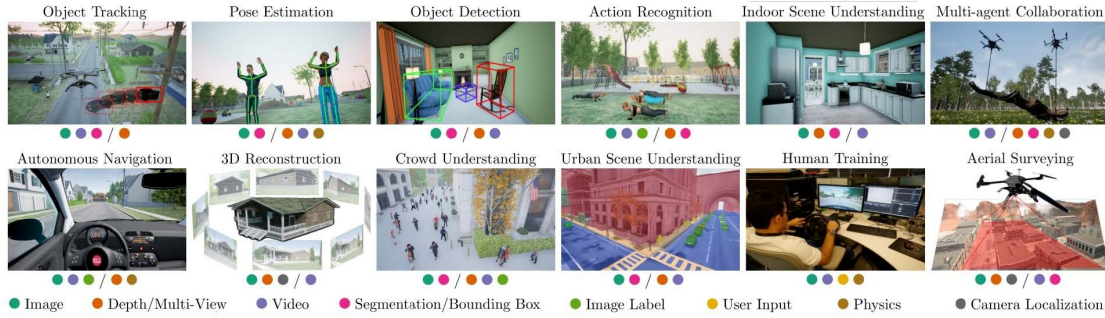


Figure 1.6: Sim4CV: A Photo-Realistic Simulator for Computer Vision Applications

Sim4CV is a simulation platform developed for research in computer vision and autonomous systems, with a strong emphasis on autonomous driving and aerial navigation [12]. Built on Unreal Engine 4, it provides high-quality visual environments and a physics engine capable of modeling realistic vehicle dynamics, object interactions, and environmental effects. The platform supports the simulation of various sensors, including cameras and lidar, enabling detailed testing of perception algorithms. Sim4CV also offers traffic and pedestrian simulation, though the level of sophistication may vary across versions and configurations (Figure 1.6). Furthermore, the ability to adjust weather and lighting conditions makes it suitable for evaluating algorithm robustness under diverse environmental scenarios.

1.3.4 Carla Simulator

The CARLA Simulator is an open-source autonomous driving platform [13] widely used in research for developing, testing, and validating autonomous vehicle algorithms. Built on Unreal Engine 4, it provides high-fidelity graphics that closely resemble real-world



Figure 1.7: Carla Simulator Screen

environments (Figure 1.7), which is essential for evaluating perception systems. CARLA’s underlying physics engine accurately models vehicle dynamics, collisions, and surface interactions, enabling realistic assessment of control and planning algorithms. It also offers comprehensive sensor simulation—including cameras, LIDAR, radar, and GNSS—allowing researchers to replicate and analyze diverse sensing conditions. In addition, CARLA supports dynamic traffic scenarios and human-like pedestrian behavior, creating complex and socially interactive environments for autonomous driving studies. Finally, its configurable weather and lighting system makes it possible to test algorithms across various environmental conditions, ensuring robustness and generalizability in real-world deployments.

Features / Simulators	Waymo Simulator	LGSVL Simulator	Sim4CV	CARLA Simulator
Graphics Quality	8/10	9/10	8/10	9/10
Physics Engine Accuracy	8/10	9/10	7/10	9/10
Sensor Simulation	8/10	9/10	7/10	9/10
Traffic and Pedestrian Simulation	8/10	9/10	6/10	9/10
Weather Conditions	7/10	8/10	6/10	9/10
Simulation at Different Times of Day	7/10	8/10	6/10	9/10

Table 1.1: Comparative Feature Scoring of Simulators

Using a set of predefined evaluation criteria—such as graphics quality, physics realism, sensor simulation, traffic modeling, and environmental variability—a comparative score was assigned to each simulator. Table 1.1 summarizes the evaluation results based on these criteria.

1.4 ADAS Applications

Advanced Driver Assistance Systems (ADAS) are becoming increasingly common in today’s vehicles. Developed to enhance vehicle safety and the driving experience, these systems utilize various technologies such as cameras, sensors, and radars to identify potential hazards and are designed to alert the driver or automatically control the vehicle under certain conditions. ADAS technologies not only assist drivers in traveling more

safely and comfortably but also have the potential to reduce traffic accidents. There are various applications of ADAS, which are crucial for the safety of the driver and also provide significant conveniences. Some of these applications can be briefly discussed as follows:

Automatic Emergency Braking (AEB):

The vehicle automatically brakes to avoid colliding with a vehicle or obstacles ahead [14]. A laser radar sensor detects the distance to the vehicle ahead and the relative speed. If there is an object in



Figure 1.8: Automatic Emergency Braking system.



Figure 1.9: Traffic Sign Recognition System

and informs the driver about traffic rules such as speed limits and prohibition signs. Generally, an image is captured by a camera sensor placed on the vehicle's windshield, and the detected image is projected onto the user's screen (Figure 1.9). This technology aims to prevent accidents by ensuring that drivers do not overlook important warning signs on the road [15].

Night Vision: This system expands the driver's field of vision in low light conditions or at night through cameras or other sensors [16]. Utilizing thermal cameras and infrared lights, it detects living beings on the road ahead and provides audible or visual warnings to the driver (Figure 1.10). In addition to detection, advanced night vision systems can highlight potential hazards directly on the dashboard display or head-up display.



Figure 1.10: Night Vision

Fatigue Detection Systems: Various studies have suggested that approximately 20% of all road accidents, and up to 50% on certain roads, are fatigue-related [17].

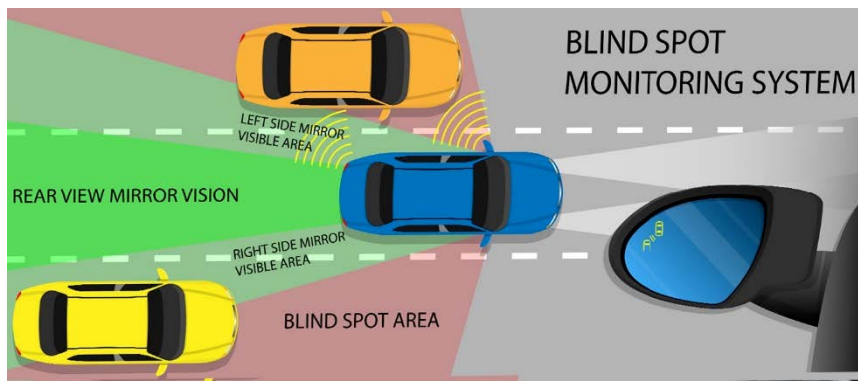


Figure 1.11: Blind Spot Warning System

Blind Spot Warning System: This system identifies the presence of other vehicles in the vehicle's blind spots and informs the driver about them. Vehicles in the blind spot are detected using radar sensors located on the sides of the rear bumper (Figure 1.11). A warning is displayed to the driver in the side mirror, and if the driver signals to change lanes while there is a vehicle in the blind spot, an alert is issued. In some models, changing lanes by braking is prevented to enhance safety of the driver in the car and to avoid road accident [18].

Parking Assistant: This feature assists the driver in better positioning the vehicle while parking and, in some cases, can automatically park the vehicle (Figure 1.12). It utilizes parking sensors located around the vehicle to do

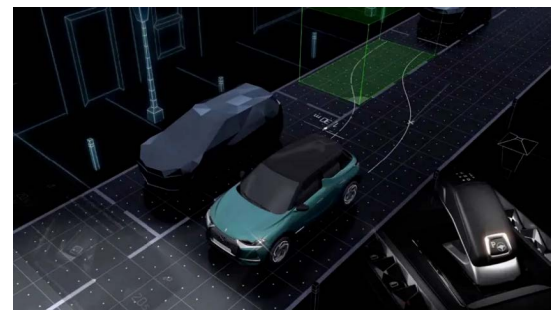
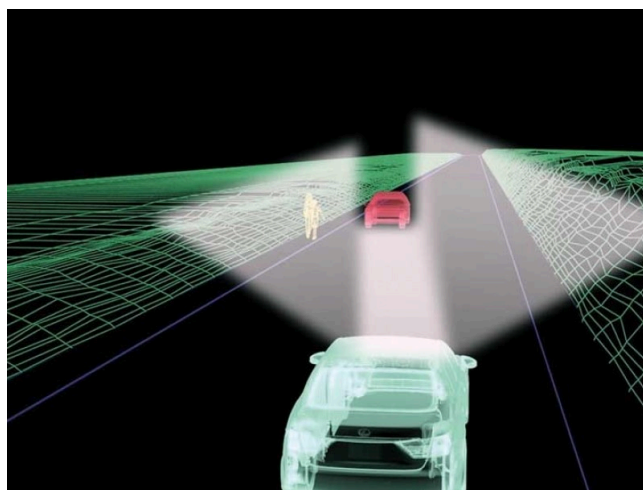


Figure 1.12: Parking Assistant



Automatic Headlight Control: The effectiveness of vehicle headlights becomes even more crucial on roads with insufficient external lighting at night, such as forest roads. Oncoming drivers on these types of roads can be affected by the headlight beams. Systems like this reduce the light intensity when detecting an oncoming vehicle to minimize the adverse effects on the opposite driver. In more advanced models, the direction of the headlight beam is adjusted to achieve this effect (Figure 1.13).

Lane Keeping Assistant: This system uses a camera sensor to detect the distance

this [19]. Parking sensors are generally electromagnetic sensors. Commonly, the system is used in a manner where the driver is still responsible for commands such as gas and gear changes.

by the headlight beams. Systems like this

of the vehicle from the lane markings and activates when the determined distance falls below a certain threshold (Figure 1.14). It steers the steering wheel to direct the vehicle back into the lane, ensuring it remains within its lane boundaries. It acts as a countermeasure to unintentional drifting caused, reducing relevant accidents by over 20%.



Figure 1.14: Lane Keeping Assistant

Adaptive Cruise Control (ACC):

This system is a more advanced version of the traditional cruise control system [20]. While a traditional cruise control maintains the vehicle at a set speed, this system adapts the speed based on the speed of the vehicle ahead (Figure 1.15). It utilizes two separate sensors to accomplish this. The first sensor is a radar sensor that measures the distance to the vehicle in front. The second sensor is a speed sensor that detects any decrease or increase in the vehicle's speed. Based on the information from these sensors, decisions are made and implemented to accelerate or decelerate. Generally, the desired distance and following speed can be adjusted by the user via controls on the steering wheel.

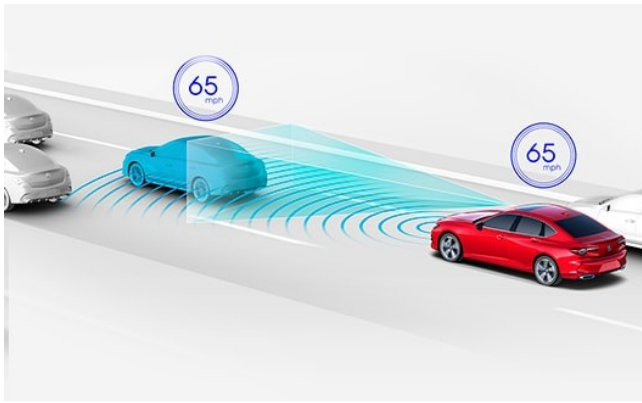


Figure 1.15: Adaptive Cruise Control

1.5 Related Works

The research community has proposed various methods for validating autonomous driving functions using simulation and Vehicle-in-the-Loop (ViL) concepts, as well as a wide range of lateral control strategies for path and lane tracking. This section summarises the most relevant works and explains how they relate to the present thesis, which implements a Lane Keeping Assist (LKA) function in the CARLA simulator using an external Python-based controller and an RGB camera.

Vehicle-in-the-Loop simulation

Son et al. propose a PG-Based Vehicle-In-the-Loop Simulation (VILS) framework that combines a real vehicle with a virtual driving environment for system development and consistency validation [21]. Their system includes modules for virtual road generation, synchronisation between real and virtual domains, a virtual traffic manager, and perception sensor modelling. They demonstrate that the behaviour observed in the VILS setup

is consistent with physical vehicle tests over different speeds, road types and surrounding environments. This work shows how ViL can improve reproducibility and safety when testing autonomous driving systems.

Cheng et al. provide a comprehensive survey on testbench-based Vehicle-in-the-Loop simulation testing for autonomous vehicles [22]. They review existing autonomous vehicle testing approaches and identify limitations of traditional methods. The paper then focuses on testbench-based ViL architectures, principles and equipment, with particular emphasis on physical signal stimulation and realistic road condition simulation. The survey identifies research gaps between current testbench technologies and industrial AV testing needs, and highlights ViL as a promising tool for accurate and efficient validation.

Xiong Hu et al. present a ViL simulation test framework for autonomous vehicles, where an autonomous driving system is evaluated using a virtual environment combined with real vehicle components [23]. Their work emphasises the synchronisation of vehicle dynamics and simulated scenarios and discusses how ViL can be used to test autonomous driving functions safely and systematically.

These works clearly show that ViL is an important methodology for validating autonomous driving systems. However, they mainly focus on general architectures, testbench setups, and system-level validation. They do not specifically address a software-only ViL configuration in a driving simulator like CARLA, where a simulated vehicle is controlled by an external code using RGB camera images. Furthermore, the detailed design and evaluation of a single ADAS function such as Lane Keeping Assist—implemented and tested entirely in such a ViL setup—remains less explored. This is where the present thesis contributes.

Lateral control for autonomous vehicles

Lateral control is a key component for lane keeping and path tracking. Kebbaty et al. provide a technical review of lateral control methods for autonomous wheeled vehicles [24]. They classify control approaches into model-based and data-driven methods, and discuss their advantages, limitations and implementation challenges. The survey underlines that lateral control must deal with accuracy, robustness and real-time constraints, and highlights open challenges for future research.

Artuñedo et al. present a comparative evaluation of state-of-the-art lateral control strategies for autonomous vehicles [25]. They propose a systematic tuning methodology and define a comprehensive set of performance metrics covering tracking accuracy, robustness and comfort. The controllers are evaluated in extensive simulations and real-vehicle tests over different trajectories and scenarios. Their results show how different control strategies behave under the same conditions and provide guidance for selecting suitable lateral controllers.

Dominguez et al. experimentally compare several classical lateral controllers—such as Pure Pursuit and Stanley—on an autonomous vehicle platform [26]. They study the performance of these controllers in terms of tracking error and stability, using a combi-

nation of simulation and real-world experiments. Their work provides practical insight into the behaviour of commonly used path-tracking controllers and their suitability for autonomous driving applications.

Snider’s technical report is a widely cited reference on automatic steering methods for autonomous automobile path tracking [27]. It describes the mathematical formulation and implementation details of several steering controllers, including Pure Pursuit and Stanley, and discusses their application to autonomous vehicles. This report is frequently used as a theoretical foundation when implementing classical lateral control laws.

These studies provide a solid theoretical and experimental basis for the lateral control part of Lane Keeping Assist. However, most of them either assume ideal path or lane information (for example, predefined waypoints or ground-truth lane geometry) or focus on real-vehicle experiments without an explicit simulation-based ViL framework. They do not analyse in detail how classical lateral controllers perform when they receive lane information from a real-time camera-based perception pipeline inside a simulator.

Identified gap and relation to this thesis

In summary, the literature on ViL shows how combined real–virtual setups can be used to validate autonomous driving systems [21],[22],[23]}, while the lateral control literature provides a variety of controllers and comparative evaluations for path and lane tracking [24],[25],[26],[27]}. However, the combination of these two aspects

- a **Vehicle-in-the-Loop-style simulation environment** in a high-fidelity simulator, and
- a **camera-based Lane Keeping Assist function** implemented as an external control stack — is less represented in the existing works.

The present thesis addresses this gap by implementing a modular LKA system in CARLA, where an RGB camera provides input to an external Python program that estimates lane geometry and computes steering commands. The system is evaluated in a reproducible simulation environment, thus connecting the ideas of ViL-based validation and classical lateral control within a single, integrated Lane Keeping Assist framework.

Chapter 2

Interface Co-Simulation Design

2.1 Selecting the Suitable Simulator

The development and testing of autonomous driving technologies require a robust simulation environment. This environment must accurately model the real world, including vehicles, pedestrians, and various environmental conditions, while also providing comprehensive support for sensor simulation and enabling integration with analytical tools. After a detailed evaluation of existing simulators, including the Waymo Simulator, LGSVL Simulator, Sim4CV, and CARLA Simulator, based on critical features such as graphic quality, the accuracy of the physics engine, sensor simulation capabilities, the simulation of traffic and pedestrians, weather conditions, the ability to simulate different times of day, CARLA Simulator has been identified as the most suitable choice for my research objectives. CARLA provides high-quality graphics with Unreal Engine 4 [28] and its physics engine accurately models vehicle dynamics and environmental interactions, offering a solid foundation for testing autonomous driving algorithms under various conditions. Moreover, its ability to simulate a wide range of sensors used in autonomous vehicles, such as cameras, LIDAR, radar, and GNSS, with high fidelity is crucial for the development and testing of perception algorithms. CARLA also excels in simulating dynamic traffic scenarios and pedestrian behaviors, facilitating comprehensive testing of autonomous driving systems in complex urban environments. The capability to simulate different weather conditions and times of day is important for assessing the performance of autonomous vehicle systems under various environmental conditions. For my research, CARLA's Python API facilitates easy integration with an external controller, providing a seamless workflow for data analysis and algorithm testing. Consequently, CARLA Simulator's advanced graphics, accurate physics engine, extensive sensor simulation capabilities, and effective traffic and pedestrian simulation set it as the ideal choice for my autonomous driving research. Its compatibility with Python API further supports my analytical and development needs, making it the most suitable simulator for my project.

2.2 Modeling the Car in the Environment

In order to control a vehicle in the simulation, the controller must recognize the vehicle model. For this, a model of the vehicle is needed. Various models can be used when modeling the vehicle. Among these, the most commonly used ones usually offer a good

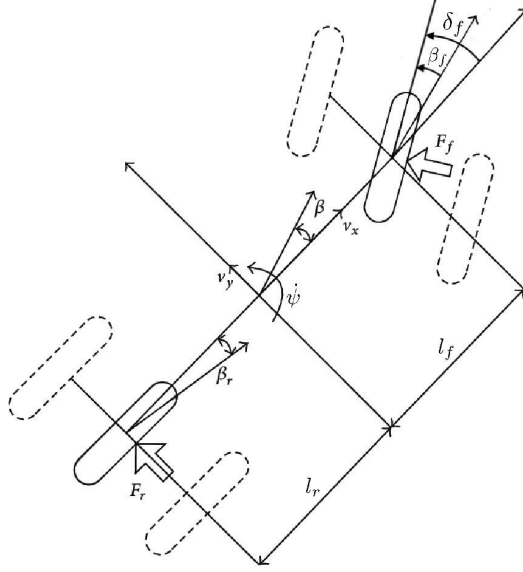


Figure 2.1: Single Track Model

balance between simplicity and accuracy, capable of representing the vehicle's motion dynamics and control systems. In line with the needs of this project, the Dynamic Single-Track (DST) model has been chosen (shown in Figure 2.1). The DST model (bicycle model) can be modeled with single-track wheels (one front and one rear wheel). This is equivalent to a model where the right and left sides of a four-wheeled vehicle are considered equal.

Vehicle variables:

- δ_f : steering angle.
- β : vehicle slip angle = angle between the vehicle longitudinal axis and velocity
- β_f, β_r : tire slip angles = angles between the tire longitudinal axis and velocity.

Vehicle parameters:

- CoG: center of gravity
- m, J : mass and moment of inertia
- l_f : distance CoG - front wheel center
- l_r : distance CoG - rear wheel center
- c_f, c_r : front/rear cornering stiffnesses.

Vehicle Dynamic parameters:

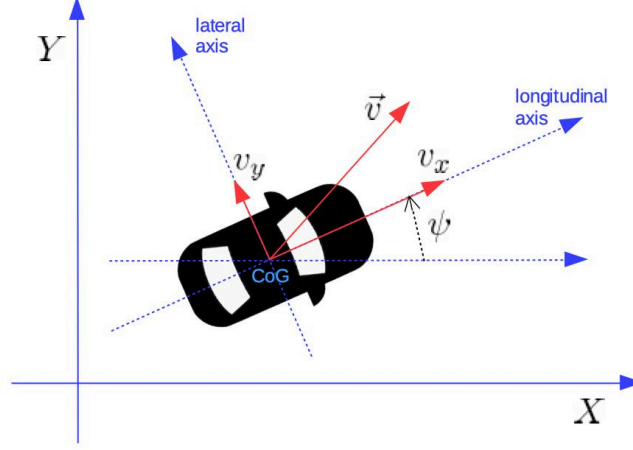


Figure 2.2: Vehicle Reference System

- X, Y : coordinates of the vehicle CoG in an inertial reference frame (shown in Figure 2.2)
- ψ : yaw angle
- $\omega_\psi = \dot{\psi}$: yaw rate
- $\vec{v} \equiv V$: velocity vector in the inertial frame
- v_x : longitudinal speed = $\vec{v}$ component along the longitudinal axis
- v_y : lateral speed = $\vec{v}$ component along the lateral (transverse) axis
- a_x : longitudinal acceleration in the inertial frame.

The state equations of the DST model are:

$$\dot{X} = V_x \cos \psi - V_y \sin \psi \quad (2.1)$$

$$\dot{Y} = V_x \sin \psi + V_y \cos \psi \quad (2.2)$$

$$\dot{\psi} = \omega_\psi \quad (2.3)$$

$$\dot{V}_x = V_y \omega_\psi + a_x \quad (2.4)$$

$$\dot{V}_y = -V_x \omega_\psi + \frac{2}{m}(F_{yf} + F_{yr}) \quad (2.5)$$

$$\dot{\omega}_\psi = \frac{2}{J}(l_f F_{yf} - l_r F_{yr}) \quad (2.6)$$

where F_{yf} and F_{yr} are the lateral forces exchanged between tire and road.

For the tire model, there are several tire model options:

Linear (for $V_x = \text{const}$) tire model:

$$F_{yf} = -C_f \beta_f, \quad F_{yr} = -C_r \beta_r \quad (2.7)$$

$$\beta_f = \frac{V_y + l_f \omega_\psi}{V_x} - \delta_f, \quad \beta_r = \frac{V_y - l_r \omega_\psi}{V_x} \quad (2.8)$$

Nonlinear simplified tire model:

$$F_{yf} = -C_f \beta_f \cos \delta_f, \quad F_{yr} = -C_r \beta_r \quad (2.9)$$

$$\beta_f = \arctan\left(\frac{V_y + l_f \omega_\psi}{V_x}\right) - \delta_f, \quad \beta_r = \arctan\left(\frac{V_y - l_r \omega_\psi}{V_x}\right) \quad (2.10)$$

Nonlinear Pacejka's tire model:

$$F_{yf} = -f_p(\beta_f) \cos \delta_f, \quad F_{yr} = -f_p(\beta_r) \quad (2.11)$$

where β_f and β_r and $f_p(\beta)$ is given by the Pacejka's magic formula. **Pacejka's magic formula:**

$$f_p(\beta) = p_1 \sin(p_2 \arctan(p_3 \beta - p_4(p_3 \beta - \arctan(p_3 \beta)))) \quad (2.12)$$

p_1 : peak value, p_2 : shape factor, p_3 : stiffness factor, p_4 : curvature factor. Linearizing this formula, we find $p_1 p_2 p_3 = C_f$ (or C_r).

In the real world conditions, these parameters are really hard to measure or estimate because they change based on the road conditions (see Figure 2.3).

The real parameters of the vehicle in the simulation environment have been found with the help of Carla's `Vehicle.physics` command.

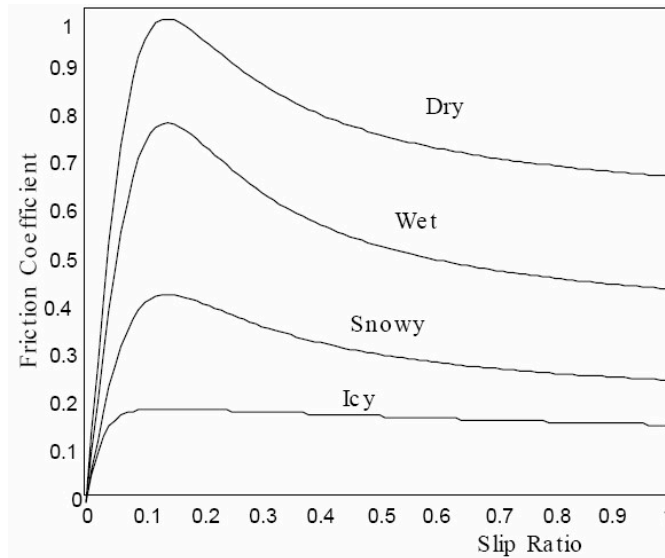


Figure 2.3: Friction Coefficients in different conditions

Vehicle Parameters Values

- m : 1318kg
- J : 2500kg/m²
- l_f : 1.54m
- l_r : 1.51m
- c_f : 15000N/rad
- c_r : 15000N/rad

2.2.1 Dispatching To Obtain Throttle-Brake Value From Acceleration Value

To control a vehicle in the Carla simulator, we need three pieces of information: steering angle, throttle, and brake commands. However, our model generates ax data. It is not possible to directly provide this data to Carla because this information only tells us about the acceleration or deceleration of the vehicle and the magnitude of these changes. Therefore, we need to scale this data to use it for throttle-brake information. To do this, we perform dispatching. In the context of control, “dispatching” usually refers to the process of distributing tasks or resources according to certain criteria or algorithms.

To accomplish this, we first need specific reference values. These reference values should be collected according to many different scenarios to ensure they are suitable for every scenario, not just one. In our case, data were collected for three different scenarios: when the vehicle completes a straight path in a sine wave pattern, when continuous braking and accelerating are performed on a straight road, and finally, when only accelerating is done and the vehicle slows down without braking due to its own weight. The data include the vehicle’s ax acceleration value, and throttle and brake values. By examining these accelerations, the priorities of throttle and brake values are determined for acceleration and deceleration situations. For instance, it is not always necessary to brake for negative acceleration values; reducing the throttle can also achieve the necessary speed reduction. Therefore, each value has its weight. Also, there is a range within which each command should be applied. These ranges can be determined using an if/else structure based on the acceleration value. The commands to be applied within these ranges are found by dividing the ax value by certain weight values. Consequently, the throttle and brake values are instantaneously derived from the ax value and provided to the vehicle in the Carla environment for application. As explained above, these values were found by analyzing reference data and through trial and error. The code for the if/else structure that creates the intervals for applying these values and commands is provided below.

2.3 Data Gathering with Autonomous Driving Mode

A reference point was selected from the Carla Map to collect reference data. The vehicle was spawned at this reference point. A simulation duration was determined based on a finish point we had predetermined. The vehicle was moved using Carla’s autonomous driving mode. During this time, the necessary reference position, waypoint data, yaw angle, and speed data were recorded into a `.csv` file at intervals of $T_s = 0.05$ seconds. Once the required simulation duration was completed, the vehicle was removed from the map.

Two collector routines are used during the recording. The first one, `carla_data_collector`, queries the full localization information of the vehicle at every tick: world position (x, y, z) , rotation (pitch, yaw, roll), linear velocity, linear acceleration and angular velocity. The second one, `carla_data_collector_2`, focuses on the geometry of the lane and returns, for each tick, the position of the closest waypoint together with the unit forward vector $\hat{t} = (t_x, t_y)$ tangent to the lane in the legal direction of travel. The two collectors are complementary: the first one captures the dynamic state of the ego vehicle, the second one captures the geometric reference that the lane-keeping controller will track. The complete listings of both routines are reported in Section B for completeness; here we only describe the meaning of the recorded fields.

In order to keep the resulting `.csv` file compact, the second collector is wrapped by a duplicate-suppression block. The position of the closest waypoint can remain unchanged for several consecutive ticks—for instance when the vehicle stops at a traffic light—and writing identical rows to the file would simply increase its size without adding information. The block compares each new sample with the last one written and appends it only if at least one of the two coordinates has changed. The resulting file therefore contains a strictly monotonic sampling of the centre line, indexed by the order in which the waypoints were visited.

2.4 Data Gathering with Manual Driving Mode

The first capability that we demonstrate with CARLA is localization, which allows our ego-vehicle to determine its pose in the world. Two coordinate frames: the map frame, which is a coordinate frame that is fixed at the initial position of the map, and the vehicle frame, which is a coordinate frame attached to the middle of the rear axle of the vehicle. For our particular experiment, we record the vehicle’s accurate pose while traversing a curved-straight route in the Town10. In the Python API, there is a class that comprises all the localization information for an actor at a certain moment in time; its methods comprise Getters such as `get_acceleration`, `get_velocity`, `get_transform`, and `get_angular_velocity`- which in our case we utilize the `get_transform` method which includes both location of the object (X, Y, Z from the origin of the map) in meters,

and its rotation characteristics from which we utilize the yaw values. More concisely, X and Y coordinates were our focus, as in the map chosen, the road is entirely flat. Hence only these two coordinates remain crucial for tracking the vehicle’s trajectory. Also, the Yaw angle, describing the orientation of the map’s coordinate system, is essential for indication of the direction the vehicle is facing. Therefore for curved paths, which our scenarios contain, it is important.

Some additional considerations also are to be noted, such as the sampling rate, which determines the time step of the data collection. We have chosen 0.1, indicating not too sparse to miss some critical dynamics, especially for sharp turns, and also not too frequent leading to redundant information.

After initializing the scenario and ego vehicle (based on the preferences), based on the duration of the scenario depending on the controls of the car (throttle, brake, and steering commands), we record the data of the vehicle at each time step. The corresponding logging routine has the same overall structure as in the autonomous mode—at every simulation tick the snapshot frame, the pose, the velocity, the acceleration and the applied control are appended to a Pandas dictionary which is finally written to disk as a `.csv` file. The full code excerpt is provided in Section B.

In manual driving mode, the autopilot is disengaged with `vehicle.set_autopilot(False)` and the user is given full control of throttle, brake, and steering through the keyboard. Each pressed key is captured as a PyGame event and is translated into a `carla.VehicleControl` object that is then applied to the ego vehicle in the simulator. While the user drives, the same getters introduced for the autonomous mode are queried at every simulation tick, so that the resulting `.csv` file shares the same structure for both modes and can be loaded by the controller without any additional preprocessing.

This dual-mode acquisition serves two purposes. First, it provides reference trajectories for the dispatcher tuning and for the controller design, since the autonomous mode of CARLA produces well-formed paths along the center of the lane. Second, it allows the recording of edge cases that are hard to reproduce automatically—for example aggressive lane changes, harsh braking, or following curves at the limit of stability—because they can be performed deliberately by a human driver. The two `.csv` files are then used as the input data set for the lane keeping assist controller described in Chapter 3.

Chapter 3

Lane Keeping Assist: Theory, Architectures and Method Comparison

The previous chapter established the simulation environment, the vehicle model and the data-gathering pipeline. This chapter develops the lateral-control component of the Lane Keeping Assist (LKA) function and presents three progressively richer architectures that have been implemented in the same CARLA framework. The discussion is deliberately kept theoretical: the goal of the chapter is to make explicit *why* each design choice was made, what the corresponding plant model looks like, and what the analytical tuning procedure predicts in terms of closed-loop behaviour. Code excerpts are gathered in Section B so that the present chapter can focus on the reasoning. The numerical evaluation of each controller in the simulator is the subject of the next chapter.

The chapter is structured as follows. Section 3.1 gives a high-level description of the three architectures and motivates the order in which they are introduced. Section 3.2 revisits the kinematic bicycle model of Section 2 from the perspective of control-oriented modelling, and derives the input-output transfer function from the steering command to the lateral state of the vehicle. Section 3.3 recalls the PID feedback law and its discrete-time implementation, with particular attention to the issues of integral wind-up and output saturation that arise in any practical realisation. Section 3.4 and Section 3.5 then specialise the general framework to the two error signals that are most commonly used in path-following applications: the cross-track distance and the heading error to a look-ahead point. The two formulations differ only in the choice of error signal, but as it will be shown, this single choice has profound consequences on the order of the resulting plant, on the type of controller that is required for stability, and on the way the gains have to be scheduled with vehicle speed. Section 3.6 presents a systematic identification procedure that fits the parameters of the plant from open-loop step responses recorded in the simulator, and derives analytical PID gains from a pole-placement criterion. Section 3.7 finally extends the heading-error formulation with a perception layer based on a deep convolutional lane-detection network and on inverse perspective mapping, which removes the dependence on a pre-recorded reference path. Section 3.8 summarises the relative merits of the three approaches from a control-theoretic standpoint.

3.1 Overview of the Three Architectures

The lateral-control function has been implemented in three increasingly sophisticated forms, each of which is studied in a dedicated module of the source tree:

- *Cross-track PID* (`lane_shift_pid/`). The lateral error is the signed perpendicular distance between the vehicle and the closest waypoint on the recorded reference path. A discrete-time PID controller produces the steering command directly from this distance.
- *Heading-error PID* (`heading_error_pid/`). The lateral error is the signed angle between the vehicle’s forward direction and the line that connects the vehicle to a look-ahead point placed on the reference path at a speed-dependent distance ahead. The error signal is now an angle, and a discrete-time PI controller is sufficient to drive it to zero.
- *Vision-based heading-error PID* (`detection&pid.py`). The reference path is no longer read from a pre-recorded `.csv` file. Instead, the centre line of the lane is reconstructed at every tick from the RGB camera using a deep convolutional lane-detection network and an inverse-perspective-mapping projection. A look-ahead target is selected on the reconstructed centre line and is fed to the same PI controller as in the previous case.

The three architectures are sketched side by side in Figure 3.1. They share the same actuator (the `carla.VehicleControl.steer` field), the same simulation rate ($T_s = 0.05$ s), the same vehicle dynamics, and—crucially—the same lateral plant up to the choice of the error signal. The differences between them are confined to two well-defined components: the way the lateral error is constructed, and the structure of the PID controller that consumes it. This modular separation makes it possible to compare the three approaches on equal footing, and to attribute the observed differences in behaviour to specific design choices rather than to incidental implementation details.

3.2 The Kinematic Bicycle Model Revisited

The kinematic bicycle model introduced in Section 2 is the simplest description of the planar motion of a four-wheeled vehicle. It collapses the front pair of wheels into a single equivalent wheel placed at the front axle, and similarly for the rear pair, and ignores tyre slip. Under these assumptions the configuration of the vehicle is fully described by three quantities: the position (x, y) of a reference point—conventionally taken at the centre of the rear axle—and the yaw angle ψ that the body of the vehicle makes with the world x

Three lateral-control architectures compared

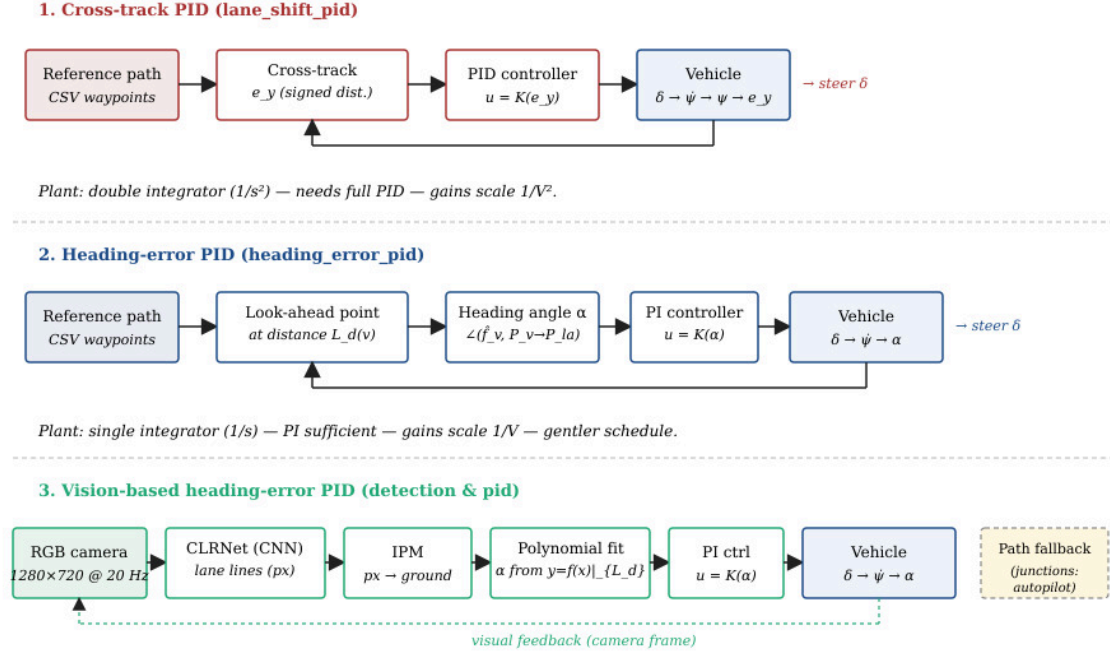


Figure 3.1: Block-diagram comparison of the three lateral-control architectures investigated in this thesis. The first one closes the loop on the cross-track distance computed against a recorded path; the second one closes the loop on a heading angle derived from a speed-dependent look-ahead point; the third one replaces the recorded path with a perception pipeline that reconstructs the centre line of the lane from the camera image. All three architectures share the same actuator, the same vehicle dynamics, and the same PID structure.

-axis. The configuration space is therefore $\mathbb{R}^2 \times S^1$, and a useful sketch of its variables is given in Figure 3.2.

Two scalar control inputs act on the model: the longitudinal speed v at the rear axle, controlled through the throttle and brake pedals, and the steering angle δ at the front wheel. With these inputs, the equations of motion of the bicycle are

$$\dot{x} = v \cos \psi \quad (3.13)$$

$$\dot{y} = v \sin \psi \quad (3.14)$$

$$\dot{\psi} = \frac{v}{L} \tan \delta \quad (3.15)$$

where L is the wheelbase. For the small steering angles that occur in normal driving—the LKA function is by definition active only inside the lane, so $|\delta| < 5^\circ$ in almost every situation—the small-angle approximation $\tan \delta \approx \delta$ holds with negligible error and the yaw-rate equation simplifies to

Kinematic bicycle model

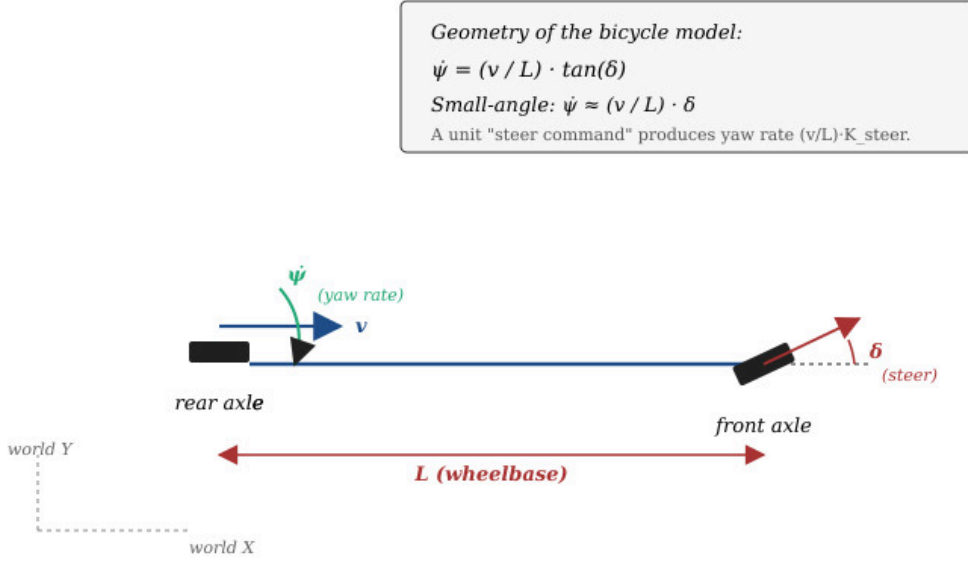


Figure 3.2: Kinematic bicycle model. The vehicle has wheelbase L , longitudinal speed v at the rear axle, yaw angle ψ relative to the world x -axis, and a front steering angle δ measured from the longitudinal axis of the body. The yaw rate $\dot{\psi}$ is determined by the geometry of the model.

$$\dot{\psi} = \frac{v}{L} \delta \quad (3.16)$$

In the CARLA Python API, the steering input is not the physical steering angle δ but a normalised “steer command” $\delta_c \in [-1, +1]$ that the simulator scales internally by a vehicle-specific maximum angle $\delta_{\max}$. The relationship $\delta = K_{\text{steer}} \delta_c$ with $K_{\text{steer}} \approx \delta_{\max}$ is approximately linear in the regime of interest [24]. Plugging this relationship into the yaw-rate equation gives the relation between the actuator command and the resulting yaw rate:

$$\dot{\psi} = v \frac{K_{\text{steer}}}{L} \delta_c \quad (3.17)$$

The key observation is that $\dot{\psi}$ depends linearly on the steering command and on the speed. The factor $\frac{v}{L}$ appears because at higher speeds the same steering angle traces a larger arc per unit time. From a control-theory perspective, the steering channel of the vehicle is an *open-loop integrator* whose gain scales with speed: if the steering command is held constant, the yaw rate is constant and the heading angle grows linearly in time. This integrator is the building block from which the two plants of Section 3.4 and Section 3.5 will be constructed.

3.3 The PID Feedback Law

A Proportional–Integral–Derivative controller computes the control action $u(t)$ as a linear combination of three terms: the present error, the accumulated past error, and the rate of change of the error. Given a scalar error signal $e(t)$, the continuous-time PID law is

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (3.18)$$

Each term plays a specific role in the LKA context. The proportional term $K_p e(t)$ produces an immediate reaction whose magnitude is proportional to the present error and is responsible for the fast component of the response. The integral term $K_i \int e(\tau) d\tau$ accumulates the past error and removes the steady-state offset that would otherwise persist when the plant has finite DC gain or when the disturbance is constant—for instance on a constantly cambered road, or under a small wheel-alignment offset. The derivative term $K_d \dot{e}(t)$ anticipates the future evolution of the error from its rate of change and adds damping when the error is approaching zero. The qualitative effect of changing each gain on the closed-loop response is summarised in Table 3.2; these dependencies are well known from the control literature [29] and are repeated here only because they directly inform the manual tuning that has been performed during the development.

Gain	Rise time	Overshoot	Settling time	Steady-state error
Increase K_p	decreases	increases	small change	decreases
Increase K_i	decreases	increases	increases	eliminates
Increase K_d	small change	decreases	decreases	no effect

Table 3.2: Qualitative effect of each PID gain on the closed-loop response of a generic plant.

3.3.1 Discrete-Time Implementation

The controller runs at the same frequency as the simulator, that is at $T_s = 0.05$ s. The continuous-time integral and derivative therefore have to be approximated. Using the backward-Euler rule for the integral and a first-order backward difference for the derivative, the PID law in Equation (3.18) becomes

$$I[k] = I[k-1] + e[k]T_s \quad (3.19)$$

$$D[k] = \frac{e[k] - e[k-1]}{T_s} \quad (3.20)$$

$$u[k] = K_p e[k] + K_i I[k] + K_d D[k] \quad (3.21)$$

In the actual implementation, the integral and the derivative are not stored as scalar variables but as the running sum and the first difference of a fixed-length deque of the most recent error samples. This windowed formulation provides a built-in saturation of

the integral term—the deque contains at most ten samples in the present implementation—and it makes the derivative term less sensitive to high-frequency noise on the error signal.

3.3.2 Anti-Windup and Saturation

Two practical issues must be addressed in any discrete realisation of Equation (3.21). The first is *integral wind-up*: when the steering command saturates at the physical limit of the actuator, the integral term keeps growing as long as the lateral error does not change sign. Once the vehicle finally crosses the centre of the lane, the controller has to “discharge” this accumulated integral before it can produce a counter-steering action, which leads to a long undershoot. The simplest remedy, used in all three architectures of the present work, is to clamp the integral to a symmetric interval $[-I_{\max}, +I_{\max}]$ at every step.

The second issue is *output saturation* proper. The CARLA `VehicleControl.steer` field accepts values in $[-1, +1]$, with -1 corresponding to a fully left and $+1$ to a fully right steering wheel. Any control law must therefore saturate its output to this range; in the implementation, the maximum value is set to 0.8 , slightly below the physical limit, in order to leave a margin for transient overshoots—a choice that matches the default of the CARLA reference local planner [30].

A third, more subtle issue is the *rate of change* of the steering command. A PID controller is intrinsically a memoryless operator on the error sequence and can produce arbitrarily large jumps between consecutive samples, especially if the derivative term is large. Such jumps would be felt by the vehicle as a step on the steering rack and would excite high-frequency suspension dynamics. To avoid this, the steering command is rate-limited at every tick to a maximum increment of 0.1 per step, which corresponds approximately to a $50^\circ/\text{s}$ steering-wheel speed at the saturation level—well within the capabilities of a human driver, and well below the bandwidth at which the simulator becomes numerically unstable.

3.4 Cross-Track Error Formulation

The first of the three architectures closes the loop on the *cross-track error* e_y , defined as the signed perpendicular distance between the vehicle and the closest waypoint on the reference path. The CARLA waypoint graph exposes, at every position of the world, the closest waypoint together with the unit forward vector $\hat{t} = (t_x, t_y)$ tangent to the lane. Given the vehicle position $P_v = (x_v, y_v)$ and the closest waypoint $P_w = (x_w, y_w)$, the cross-track error is the projection of the displacement $\vec{r} = P_w - P_v$ on the unit normal $\hat{n} = (-t_y, t_x)$:

$$e_y = \vec{r} \cdot \hat{n} = (x_w - x_v)(-t_y) + (y_w - y_v)t_x \quad (3.22)$$

Lane-shift / cross-track geometry

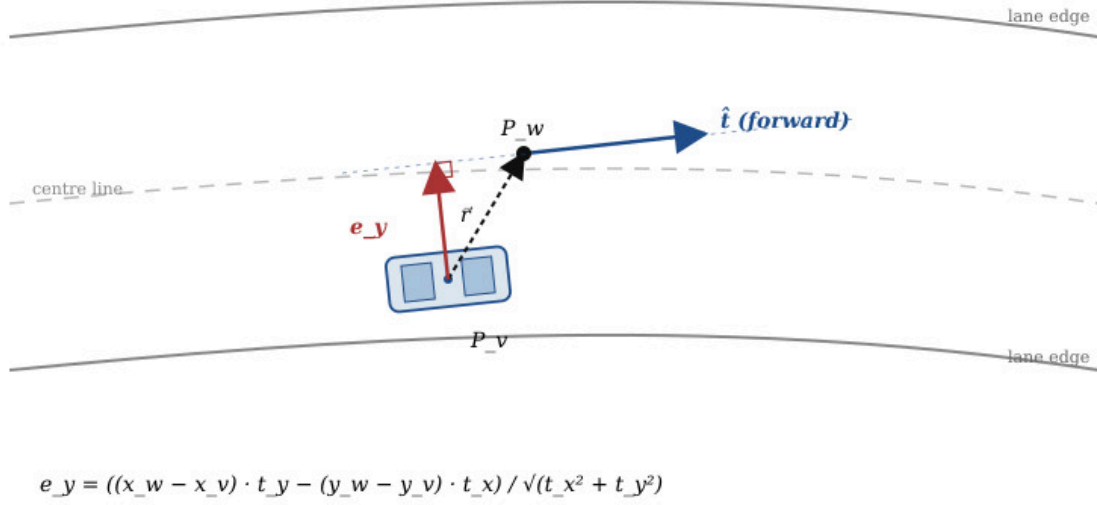


Figure 3.3: Geometric definition of the cross-track error e_y as the signed perpendicular distance between the vehicle position P_v and the line through the closest waypoint P_w in the direction of the lane forward vector $\hat{t}$. Positive values of e_y correspond to the vehicle being on the right of the lane centre with respect to the direction of travel. which, after rearrangement of the signs and division by the norm of $\hat{t}$ in case the forward vector returned by the API is not exactly normalised, becomes

$$e_y = \frac{(x_w - x_v)t_y - (y_w - y_v)t_x}{\sqrt{t_x^2 + t_y^2}} \quad (3.23)$$

The geometric construction has been illustrated in Figure 3.3 and is not repeated here.

3.4.1 The Cross-Track Plant

To derive the transfer function from the steering command to the cross-track error, three simple integrations have to be chained. Starting from the bicycle-model relation $\dot{\psi} = vK_{\text{steer}} \frac{\delta_c}{L}$ established in Section 3.2:

- The yaw rate $\dot{\psi}$ integrated once gives the heading deviation ψ from the path tangent.
- The heading deviation ψ multiplied by the vehicle speed v gives the lateral velocity $\dot{y}$ relative to the path (small-angle approximation).
- The lateral velocity $\dot{y}$ integrated once gives the cross-track distance e_y .

In the Laplace domain, the chain of operations is

$$\frac{E_{y(s)}}{\Delta_c(s)} = \frac{vK_{\text{steer}}}{L} \cdot \frac{1}{s} \cdot v \cdot \frac{1}{s} = \frac{v^2 K_{\text{steer}}}{Ls^2} \quad (3.24)$$

The plant from the steering command to the cross-track error is therefore a *double integrator*, with a DC gain that scales with v^2 . This is the central observation that drives every other property of the cross-track formulation. A double integrator is marginally stable on its own—it has two poles at the origin—and any feedback law that uses only the proportional and integral terms produces a closed-loop system with poles either on the imaginary axis or in the right half-plane. The derivative term is therefore *necessary* for stability, not optional. Furthermore, the v^2 scaling of the DC gain means that, in order to keep the closed-loop bandwidth constant as the vehicle accelerates, all three PID gains have to be re-scaled by $\frac{1}{v^2}$. This is a steep dependence that will be revisited in Section 3.6.

3.4.2 Pole Placement

A natural choice for the closed-loop denominator of a PID compensating a double integrator is a triple real pole at $s = -\omega_n$, which gives a critically damped behaviour with bandwidth ω_n . The closed-loop characteristic polynomial of $K_p + \frac{K_i}{s} + K_d s$ acting on $\frac{K_{\text{lat}}}{s^2}$ with $K_{\text{lat}} = v^2 \frac{K_{\text{steer}}}{L}$ is

$$s^3 + (K_{\text{lat}} K_d) s^2 + (K_{\text{lat}} K_p) s + K_{\text{lat}} K_i \quad (3.25)$$

Matching with $(s + \omega_n)^3 = s^3 + 3\omega_n s^2 + 3\omega_n^2 s + \omega_n^3$ yields the analytical gains

$$K_p = \frac{3\omega_n^2}{K_{\text{lat}}}, \quad K_i = \frac{\omega_n^3}{K_{\text{lat}}}, \quad K_d = \frac{3\omega_n}{K_{\text{lat}}} \quad (3.26)$$

These expressions tell, at a glance, how each gain depends on the vehicle speed: K_d scales as $\frac{1}{v^2}$ through K_{lat} , K_p does the same, and K_i —the most “expensive” gain in terms of stability margin—decreases as $v^{\{-2\}}$ as well. The tuning script `pid_tuning.py` implements exactly Equation (3.26) and produces, given a measurement of K_{steer} from the system identification of Section 3.6, the numerical gains that have been used in the simulator.

3.5 Heading-Error Formulation

The second architecture replaces the cross-track distance with a different scalar error signal: the angle between the vehicle’s forward direction and the line connecting the vehicle to a *look-ahead point* placed on the reference path at a speed-dependent distance L_d ahead of the vehicle. This formulation is closely related to the Pure Pursuit [27] and Stanley [26] geometric trackers, with the difference that here the steering command is computed by a feedback PID rather than by a fixed geometric formula. The geometric construction is illustrated in Figure 3.4.

Heading-error / look-ahead geometry

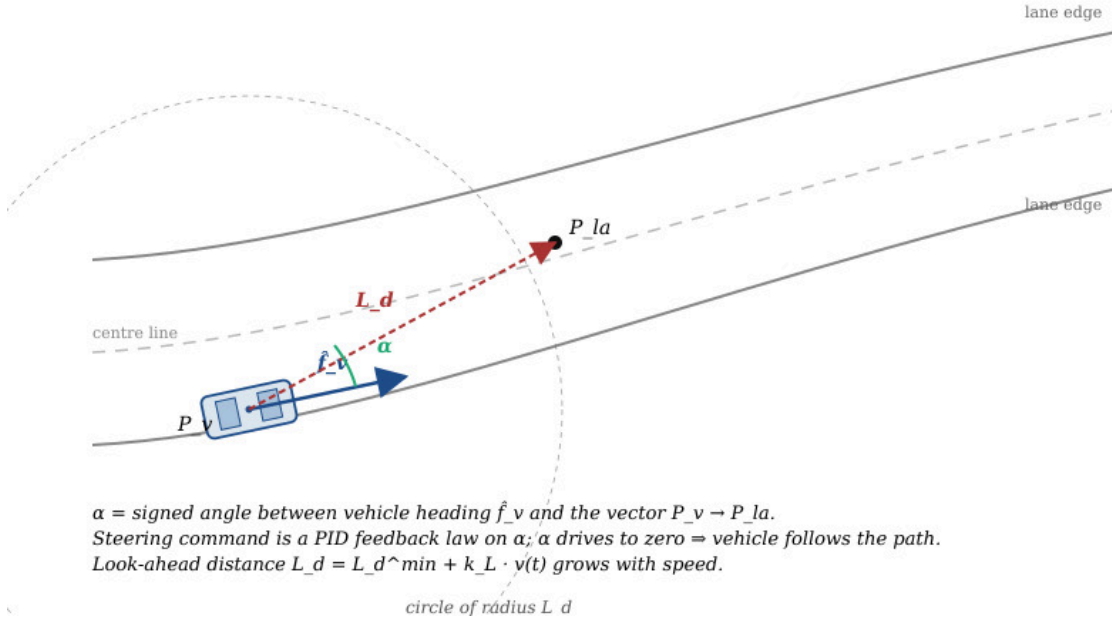


Figure 3.4: Heading-error / look-ahead geometry. The look-ahead point P_{la} is the closest waypoint on the reference path at distance at least L_d ahead of the vehicle; the heading error α is the signed angle between the vehicle forward unit vector $\hat{f}_v$ and the line $P_v \rightarrow P_{la}$. The look-ahead distance is scheduled as $L_d = L_d^{\min} + k_L v(t)$.

Formally, given the vehicle position $P_v = (x_v, y_v)$, its forward unit vector $\hat{f}_v = (f_x, f_y)$, and the look-ahead point $P_{la} = (x_l, y_l)$, the heading error α is computed from the dot product and the cross product of $\hat{f}_v$ with the displacement $\vec{d} = P_{la} - P_v$:

$$\alpha = \text{atan2}(\hat{f}_v \times \vec{d}, \hat{f}_v \cdot \vec{d}) \quad (3.27)$$

The two-argument arctangent returns a signed angle in $(-\pi, +\pi]$ and naturally handles all four quadrants. By convention—the same one used in the CARLA `agents.navigation.controller.PIDLateralController` and matched in the implementation—a positive α corresponds to a target on the right of the vehicle, which calls for a positive (right) steering command.

3.5.1 The Heading-Error Plant

Compared with the cross-track formulation, the heading-error plant has *one fewer integration*. Starting again from $\dot{\psi} = v K_{\text{steer}} \frac{\delta_c}{L}$, only one step is needed to obtain the heading deviation α relative to the look-ahead point. The look-ahead point is, by construction, ahead of the vehicle on the path; if the path is locally straight, the line to the look-ahead point is parallel to the path tangent, and the heading error reduces to the angle between the vehicle's forward vector and the path tangent. In the Laplace domain,

$$\frac{A(s)}{\Delta_{c(s)}} = \frac{vK_{\text{steer}}}{L} \cdot \frac{1}{s} = \frac{vK_{\text{steer}}}{Ls} \quad (3.28)$$

which is a *single integrator* with a DC gain that scales linearly with v , not with v^2 .

The structural difference between Equation (3.24) and Equation (3.28) is the single most important property of the present chapter, because it has three direct consequences:

- A PI controller is sufficient for closed-loop stability. The derivative term is no longer necessary—the plant has only one free integrator—and is included only if the error signal is noisy enough to require derivative filtering.
- The gains scale as $\frac{1}{v}$ instead of $\frac{1}{v^2}$. Gain scheduling across the operating speed range is therefore much gentler.
- The closed-loop bandwidth that the controller can achieve at fixed gains is higher, because each additional integrator in the plant adds 90° of phase lag and reduces the available phase margin.

The plant comparison is summarised in Figure 3.5, which puts the two transfer functions side by side and makes the role of the look-ahead point explicit: by closing the loop on the angle to a point on the path rather than on the perpendicular distance to it, the controller “consumes” one of the two integrators that the cross-track formulation has to handle.

3.5.2 Pole Placement

With the plant given by Equation (3.28) and a PI controller $C(s) = K_p + \frac{K_i}{s}$, the closed-loop characteristic polynomial reads

$$s^2 + K_{\text{lat}}K_p s + K_{\text{lat}}K_i, \quad K_{\text{lat}} = v \frac{K_{\text{steer}}}{L} \quad (3.29)$$

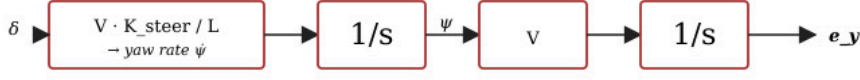
Matching with $s^2 + 2\zeta\omega_n s + \omega_n^2$ and choosing the critical damping $\zeta = 1$ yields

$$K_p = \frac{2\omega_n}{K_{\text{lat}}}, \quad K_i = \frac{\omega_n^2}{K_{\text{lat}}} \quad (3.30)$$

Both gains scale as $\frac{1}{v}$ through K_{lat} . The asymptotic comparison with the cross-track formulation is best read graphically: Figure 3.6 shows, in the bottom-right panel, the proportional gain K_p as a function of vehicle speed for the two formulations, normalised so that the value at 30 km/h is unity. The cross-track curve falls by an order of magnitude when the speed grows from 30 to 120 km/h, while the heading-error curve only falls by a factor of four. The practical consequence is that the same heading-error PI controller can be safely operated, with at most a mild gain re-scheduling, across the entire highway-speed range; the cross-track PID, in contrast, is much more sensitive to speed mismatches and would oscillate or diverge if used outside the range it has been tuned for.

Plant model comparison: cross-track vs. heading-error formulation

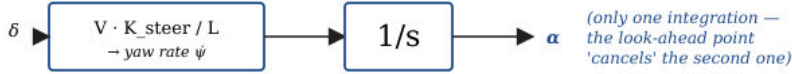
A. Cross-track error e_y formulation



Combined: $G_y(s) = (V^2 \cdot K_{steer} / L) / s^2 \rightarrow$ **double integrator**

- Plant gain scales with V^2 — gains must scale $1/V^2$ to maintain bandwidth.
- PID needed: D-term essential for stability (two free integrators).
- Pole placement: $(s + \omega_n)^3 = K_p = 3\omega_n^2/K_{lat}$, $K_i = \omega_n^3/K_{lat}$, $K_d = 3\omega_n/K_{lat}$.

B. Heading-error α formulation



Combined: $G_\alpha(s) = (V \cdot K_{steer} / L) / s \rightarrow$ **single integrator**

- Plant gain scales linearly with V — gains scale $1/V$ (gentler).
- PI is sufficient for stability; D-term optional, only for noise filtering.
- Pole placement: $(s + \omega_n)^2 = K_p = 2\zeta\omega_n/K_{lat}$, $K_i = \omega_n^2/K_{lat}$.

The choice of error signal (e_y or α) determines the plant order; the plant order determines what controller is needed.

Figure 3.5: Block-diagram comparison of the cross-track and heading-error plants. The cross-track formulation contains two integrators—yaw rate to heading, and lateral velocity to cross-track distance—while the heading-error formulation contains only one. The reduction in plant order is the reason why a PI controller is sufficient for the heading-error case, while a full PID is required for the cross-track case.

3.5.3 Look-Ahead Distance Scheduling

The look-ahead point introduces a single design parameter, the look-ahead distance L_d . A short L_d produces a controller that reacts strongly to local geometry and that is well-behaved on tight turns but easily destabilised at high speed; a long L_d produces a smooth controller that filters out high-frequency disturbances on the reference path but cuts corners on tight turns. The compromise that is most often adopted in the literature on Pure Pursuit—and that has been retained in the present implementation—is to schedule L_d linearly with the speed of the vehicle:

$$L_{d(v)} = L_d^{\min} + k_L v \quad (3.31)$$

with a minimum $L_d^{\min} = 4$ m that protects against pathological behaviour at standstill, and a slope $k_L = 0.6$ s that produces $L_d \approx 9$ m at the nominal cruise speed of 30 km/h. The same kind of linear scheduling is used by the CARLA reference local planner [30] for its waypoint-popping criterion, with comparable numerical values.

3.5.4 Heading-Error Filtering

A practical consideration that is specific to the heading-error formulation is the noise on the error signal. The look-ahead point is selected from a discrete sampling of the

reference path, so the heading error is intrinsically a piecewise-constant signal that exhibits small jumps every time the look-ahead point advances by one waypoint. These jumps are amplified by the derivative term and would produce visible jitter on the steering command. In the implementation, the heading error is therefore low-pass filtered with a first-order infinite impulse response filter of pole α_H :

$$\tilde{\alpha}[k] = \alpha_H \alpha[k] + (1 - \alpha_H) \tilde{\alpha}[k - 1] \quad (3.32)$$

with $\alpha_H = 0.4$, which corresponds to a cut-off frequency of approximately 2 Hz at the simulator rate of 20 Hz. This is well below the Nyquist frequency of the controller and well above the bandwidth of the closed loop, so it removes the per-sample jitter without affecting the dynamic response of the system.

3.6 System Identification and Analytical Tuning

The pole-placement formulas Equation (3.26) and Equation (3.30) express the PID gains in terms of two physical parameters: the wheelbase L , which is read from the vehicle blueprint through the CARLA Python API, and the steering gain K_{steer} , which depends on the specific vehicle model and is not exposed by the API. The latter has to be measured experimentally. In addition, the longitudinal channel of the LKA function—the speed regulator that maintains the cruise speed—has its own parameters that have to be identified.

The identification procedure has been implemented in the standalone script `carla_sysid.py`; it consists of two open-loop step experiments that are recorded with the simulator in synchronous mode at $T_s = 0.05$ s.

3.6.1 Longitudinal Step Experiment

The first experiment fits the parameters of a first-order longitudinal model. With the vehicle initially at rest on a long straight road, the throttle is stepped from zero to a fixed value $u_{\text{step}} = 0.5$ and held for ten seconds. The resulting speed trajectory $v(t)$ is recorded and fitted, by nonlinear least squares, to the first-order step response

$$v(t) = K u_{\text{step}} \left(1 - e^{\{-\frac{t}{\tau}\}} \right) \quad (3.33)$$

where K is the DC gain (in km/h per unit of throttle) and τ is the time constant of the longitudinal dynamics. The model is sufficient because at the moderate accelerations encountered during the experiments, the dominant non-linearity of the powertrain—the throttle-to-torque map—is approximately linear, and the slower aerodynamic drag becomes important only above 80 km/h.

3.6.2 Lateral Step Experiment

The second experiment identifies the steering gain K_{steer} . The vehicle is settled at a constant speed $V = 30$ km/h with a simple proportional speed regulator, and a small steering step $\delta_c = 0.05$ is applied for 2.5 s. The yaw rate of the vehicle is recorded directly from the `get_angular_velocity` getter of the CARLA Python API. The steady-state yaw rate $\dot{\psi}_{\text{ss}}$ obtained by averaging the last 25% of the trace is then matched against the bicycle-model prediction $\dot{\psi}_{\text{ss}} = (\frac{V}{L})K_{\text{steer}}\delta_c$, giving

$$K_{\text{steer}} = \frac{\dot{\psi}_{\text{ss}} L}{V \delta_c} \quad (3.34)$$

The duration of the lateral step is kept deliberately short, because once the vehicle starts to drift laterally the cross-track error grows quickly and the small-angle assumptions of the bicycle model are no longer satisfied.

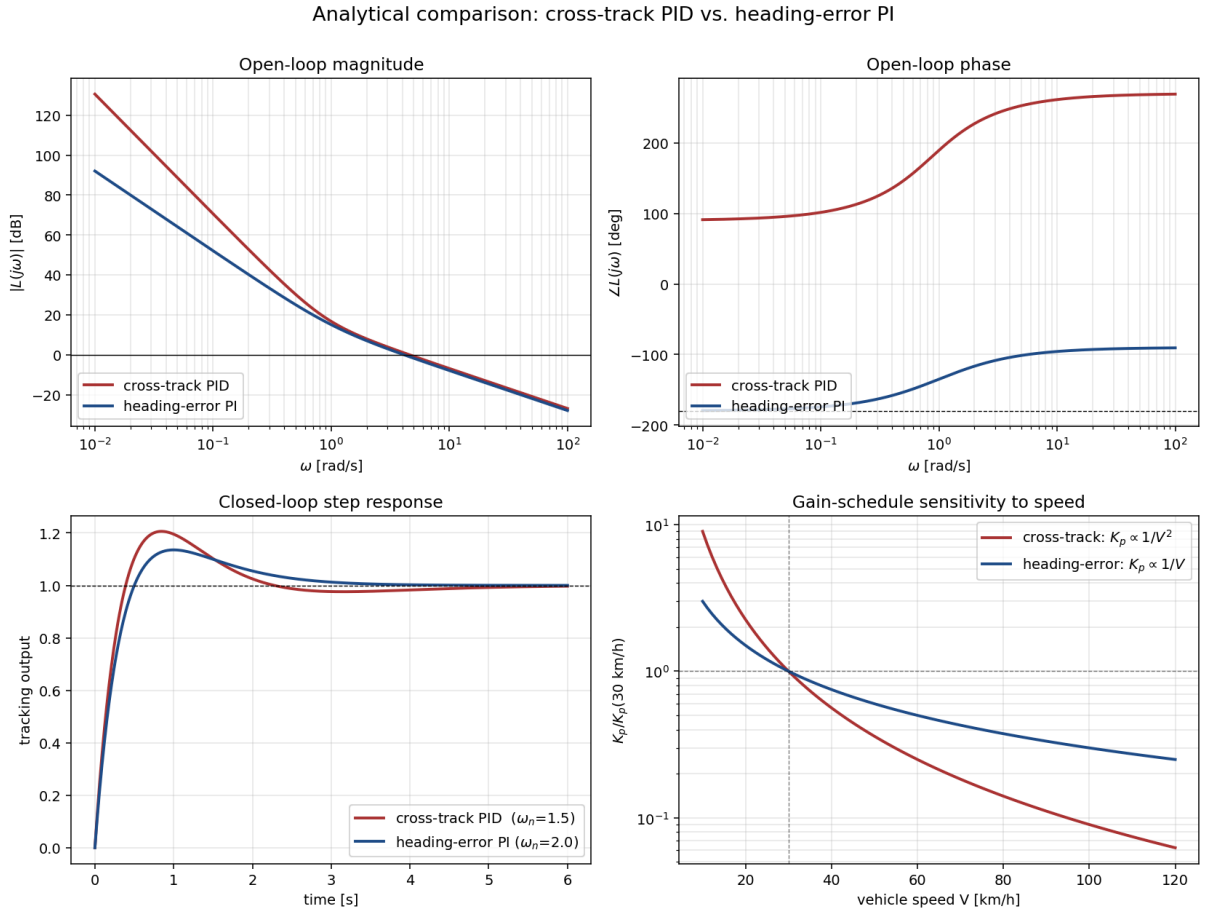


Figure 3.6: Analytical comparison of the cross-track PID and heading-error PI controllers, computed symbolically from the identified plant parameters. *Top*: open-loop magnitude and phase. *Bottom-left*: closed-loop step response. *Bottom-right*: proportional gain as a function of vehicle speed, normalised at 30 km/h. The gentler $\frac{1}{v}$ scaling of the heading-error gain, compared with the $\frac{1}{v^2}$ scaling of the cross-track gain, is the main reason why the heading-error formulation is preferred in the operational range of an LKA function.

3.6.3 Analytical Plot Comparison

With L and K_{steer} identified, the analytical gains of Equation (3.26) and Equation (3.30) can be evaluated and the resulting closed-loop transfer functions can be analysed without running the simulator. Figure 3.6 shows the four canonical control plots that the tuning script `pid_tuning.py` produces from the identified plant parameters: the open-loop magnitude and phase of the loop transfer $L(s) = C(s)G(s)$ for the two formulations (top row), the closed-loop step response of the reference-to-output transfer $T(s)$ (bottom-left), and the speed dependence of the proportional gain K_p (bottom-right).

Several observations can be made from these plots. The two open-loop magnitudes coincide above $\omega \approx 4$ rad/s—at high frequency, both loops are dominated by the fastest pole of the controller—while at low frequency the cross-track loop has a steeper slope that reflects the additional integrator. The phase of the cross-track loop reaches -180° around the same frequency at which the heading-error loop is still well above -90° , which translates into a smaller phase margin and a more oscillatory step response. Both controllers track a unit step, but the cross-track design exhibits a larger overshoot and a longer settling time, even though its bandwidth ω_n has been deliberately set lower than that of the heading-error controller to compensate for the additional plant order.

The bottom-right panel quantifies the gain-scheduling argument that has been made in Section 3.5. At $V = 120$ km/h, the proportional gain of the cross-track controller is more than ten times smaller than its value at 30 km/h, while the same gain of the heading-error controller is only four times smaller. In a fixed-gain implementation, this means that a cross-track PID tuned at 30 km/h is unsafe to use even at 50 km/h, while a heading-error PI tuned at the same speed is acceptable for the entire range 20–60 km/h that covers urban driving and most of the operational design domain of an LKA function.

3.7 Vision-Based Lane Reference

The third architecture removes the dependence on the pre-recorded `.csv` reference path by reconstructing the centre line of the lane at every tick from the RGB camera feed. The architecture is identical to the heading-error PI controller of Section 3.5 from the controller block downward; the only modification is the way in which the look-ahead point is generated. A high-level view of the vision-based pipeline is given on the bottom row of Figure 3.1.

3.7.1 The Lane-Detection Network

Lane detection in road scenes has been a benchmark task in computer vision for over a decade, and the literature offers a variety of architectures with different trade-offs between accuracy, speed, and complexity. The taxonomy adopted in the recent comparative study of Lin et al. [31] covers semantic-segmentation networks (SCNN [32], RESA [33]),

row-based classifiers (UFLD [34]), anchor-based detectors (LaneATT [35], ADNet [36]), parametric-curve regressors (BezierLaneNet [37]) and the recent CLRNet [38] family that combines an anchor-based proposal mechanism with row-wise refinement. The implementation supports any model from this list through a pluggable registry stored in the `detection_models` dictionary; the registry, the model-loading helpers, and the inference pipeline used in this thesis are built on top of the open-source UnLaneDet framework [39], which provides a unified PyTorch implementation of the architectures listed above and from which the lane-detection backbone of the present LKA controller has been adapted. For the experiments reported in this thesis, the CLRNet model trained on the TuSimple dataset has been used, because it offers the best trade-off between latency and lane-fitting accuracy in the comparative study cited above.

The detection network operates on a 1280×720 frame extracted from the camera sensor and produces a list of lane lines, each represented as a sequence of pixel coordinates (u_i, v_i) . In the present formulation, the reference for the controller is not a single lane line but the *centre line* of the ego-lane, which has to be reconstructed from the detected boundaries. The reconstruction is performed by the helper function `get_centerline`: only the lane lines that reach the bottom 30% of the image are kept—so that adjacent lanes and far-away lines are filtered out—and the closest left and right boundaries to the image centre are paired up. Their interpolated mid-points at a regular sampling along the image v -axis form the centre line, which is returned ordered from the closest sample to the furthest.

3.7.2 Inverse Perspective Mapping

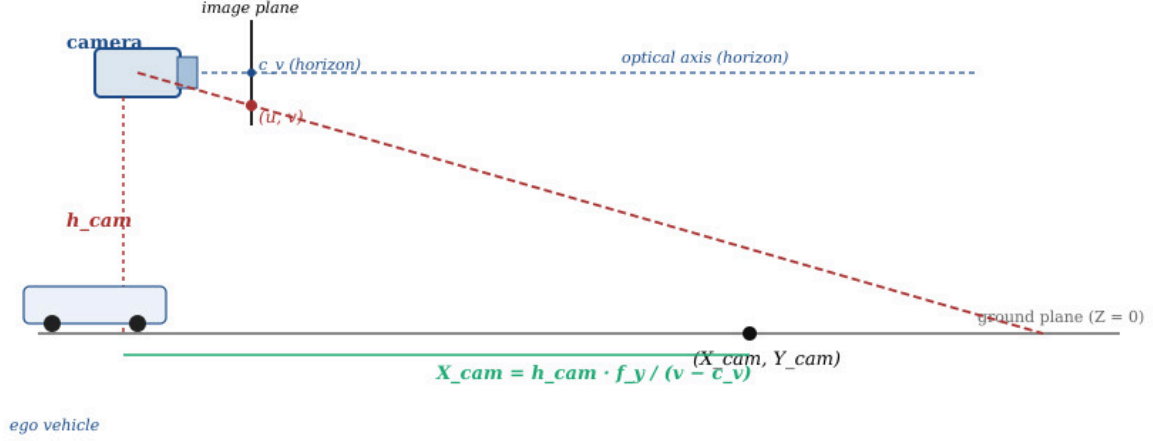
The centre line returned by the lane-detection network is expressed in pixel coordinates and cannot be used directly by the controller, which works in the body frame of the vehicle. The conversion from pixels to a body-frame position is performed by the standard *inverse perspective mapping* (IPM) projection, sketched in Figure 3.7.

Under the assumptions of a pinhole camera with zero pitch, zero roll, zero yaw, and a flat ground plane, every pixel below the horizon corresponds to a unique point on the ground plane. With the camera mounted at height h_{cam} above the road, with focal lengths $f_x = f_y = \frac{w}{2 \tan(\frac{\text{FOV}}{2})}$ derived from the image width w and the field of view, and with the principal point at $(c_u, c_v) = (\frac{w}{2}, \frac{h}{2})$, the mapping reads

$$X_{\text{cam}} = \frac{h_{\text{cam}} f_y}{v - c_v}, \quad Y_{\text{cam}} = X_{\text{cam}} \frac{u - c_u}{f_x} \quad (3.35)$$

where X_{cam} is the longitudinal distance ahead of the camera, in metres, and Y_{cam} is the lateral distance to the right of the camera. The projection is well defined only for pixels below the horizon, that is for $v > c_v$; pixels above the horizon are discarded. A maximum range $X_{\text{cam}} \leq 40$ m is also enforced, because at larger distances the small angular resolution of the pixel produces unstable lateral estimates. The constant offset of

Inverse Perspective Mapping (side view)



A pixel (u, v) below the horizon back-projects to a unique point on the ground plane.
 Lateral coordinate $Y_{cam} = X_{cam} \cdot (u - c_u) / f_x$; assumes flat ground, zero pitch and roll.
 Pixels above the horizon ($v \leq c_v$) are discarded — they project to infinity.

Figure 3.7: Inverse perspective mapping (side view). A pixel below the horizon back-projects to a unique point on the ground plane, given the camera height h_{cam} and the camera intrinsics (f_x, f_y, c_u, c_v) . Pixels above the horizon $v \leq c_v$ project to infinity and are discarded.

the camera with respect to the rear axle of the vehicle is taken into account by adding $0.8b_x$ to X_{cam} , where b_x is the half-extent of the bounding box of the vehicle.

The output of the IPM block is therefore a sampling of the centre line of the lane in the body frame of the vehicle, in metres, ordered from the closest sample to the furthest. This is exactly the input that the heading-error formulation of Section 3.5 expects, with the only difference that the look-ahead point is now selected on a perception-based reference rather than on a pre-recorded one.

3.7.3 Polynomial Fitting and Look-Ahead Selection

The samples of the centre line returned by the IPM block are noisy and unevenly spaced. To produce a stable look-ahead point, a low-degree polynomial $y = a_1x + a_0$ is fitted by least squares to the samples that fall in the longitudinal interval $[0.5, 20]$ m, and the look-ahead lateral coordinate is evaluated as $y_{la} = a_1L_d + a_0$. A first-degree polynomial—a straight line—has been adopted because the operational scenario consists of mostly straight roads and gentle curves at moderate speed; for higher speeds and tighter curves, a second-degree polynomial would be necessary, with a corresponding increase in sensitivity to outliers. The heading error is then computed as

$$\alpha = \text{atan2}(y_{la}, L_d) \quad (3.36)$$

which is the body-frame counterpart of the world-frame definition given in Section 3.5. The same PI controller, with the same low-pass filter on α , then produces the steering command.

3.7.4 Junction Handling

A specific issue that arises with the vision-based formulation, but not with the path-based formulations, is the disappearance of the lane markings inside an intersection. CARLA, like real-world road infrastructure, does not paint lane markings inside junctions, so the lane-detection network either returns nothing or returns lines that belong to the entry and exit lanes of different branches. To prevent the controller from acting on this unreliable input, a junction-detection mechanism has been added on top of the perception pipeline. At every tick, the closest waypoint is queried with `is_junction`; if either this flag is set, or the vehicle is inside one of three pre-flagged buffer regions where the lane markings are known to disappear before `is_junction` flips, the LKA function is temporarily disabled and the control is handed off to the CARLA Traffic Manager autopilot until the vehicle exits the junction. A reset of the integral state of the PI controller and of the look-ahead-distance filter is performed at every re-engagement, so that no transient is propagated from one intersection to the next.

3.8 Method Comparison

The three architectures presented in this chapter solve the same lane-keeping problem with progressively richer information and progressively more sophisticated control structure. From a purely control-theoretic standpoint, they can be summarised as in Table 3.3.

Property	Cross-track PID	Heading-error PI	Vision-based PI
Reference signal	recorded path	recorded path	reconstructed centre line
Error signal	perpendicular distance e_y	look-ahead angle α	look-ahead angle α
Plant order	2 (double integrator)	1 (single integrator)	1 (single integrator)
Required terms	$P + I + D$	$P + I$ (D optional)	$P + I$ (D optional)
Plant gain scaling	$\propto v^2$	$\propto v$	$\propto v$
Gain scheduling	$\propto 1/v^2$	$\propto 1/v$	$\propto 1/v$
Phase margin	smaller	larger	larger
Robustness to noise	moderate	good	depends on detector
Map dependence	yes (pre-recorded)	yes (pre-recorded)	no
Generalisation	restricted to recorded route	restricted to recorded route	unrestricted within ODD

Table 3.3: Side-by-side comparison of the three lateral-control architectures along the dimensions that are most relevant for an LKA function.

The progression from one architecture to the next is informed by a clear control-theoretic rationale rather than by an incremental engineering choice. The move from the cross-track to the heading-error formulation is motivated by the reduction in plant order: by closing the loop on the angle to a look-ahead point rather than on the perpendicular distance to the path, one of the two integrators of the cross-track plant is removed, the derivative term becomes optional, the gain schedule becomes gentler, and the phase margin becomes larger at any fixed bandwidth. The move from the path-based heading-error formulation to the vision-based one is motivated by the removal of the dependence on a pre-recorded reference: the same controller can be used on any road that the lane-detection network has been trained for, at the cost of an additional perception layer that has its own failure modes—most notably inside intersections, which has motivated the introduction of an explicit junction-handling logic.

The numerical evaluation of the three architectures on a common test scenario is the subject of the next chapter, which will quantify the qualitative arguments developed here in terms of cross-track error, lateral RMSE, and steering-command smoothness.

Chapter 4

Simulation Results and Evaluation

The previous chapter developed three lateral-control architectures for the Lane Keeping Assist function and motivated each design choice on control-theoretic grounds. The present chapter complements that theoretical discussion with the numerical evaluation of the three controllers in CARLA. All experiments are run on the same test route in Town06, with the same vehicle (`vehicle.nissan.patrol`), the same target speed of 30 km/h, the same simulation rate $T_s = 0.05$ s, and the same set of PID gains obtained from the system-identification procedure of Section 3.6. Under these conditions, any observed difference between the trajectories can be attributed to the structure of the error signal and to the corresponding controller rather than to incidental variations in the simulator.

The chapter is structured as follows. Section 4.1 describes the experimental protocol, the reference path and the metrics used to compare the controllers. Section 4.2 presents a qualitative view of the test route as rendered inside the CARLA world. Section 4.3 quantifies the tracking performance over the entire route. Section 4.4 and Section 4.5 then zoom in on two representative sections of the route—the long smooth curve and a sudden lane discontinuity—where the differences between the three architectures become most visible. Section 4.6 finally interprets the results in the light of the control-theoretic predictions of Section 3.

4.1 Experimental Setup

The reference path was recorded once, in autopilot mode, by driving the ego vehicle through Town06 and logging the closest waypoint at every simulation tick, as described in Section 2. The resulting `carla_data.csv` file contains a strictly monotonic sampling of the centre line along an approximately *L*-shaped route of about 850 m. The route starts on a north–south urban street, executes a 90° left turn onto a long east–west arterial, and ends after a stretch of straight road that includes a step-like lane discontinuity—a feature that proves particularly informative when comparing the three controllers. The route was selected because it exercises both the steady-state behaviour of the controllers, on the long straight section that follows the turn, and their transient behaviour, on the corner itself and on the lane discontinuity.

Three closed-loop experiments were then performed on this same route, one for each architecture:

- *Cross-track PID* (`lane_shift_pid/pygame_controller.py`). The reference is the recorded path; the error is the cross-track distance Equation (3.23); the controller is a full PID with the gains K_p, K_i, K_d produced by `pid_tuning.py` for $\omega_n^{\text{lat}} = 1.5$ rad/s. The trajectory is logged to `ego_trajectory_shift.csv` and appears in blue on every plot of this chapter.
- *Heading-error PI* (`heading_error_pid/pygame_controller.py`). The reference is again the recorded path, but the error is the look-ahead heading angle; the controller is a PI with the gains produced by `pid_tuning.py` for $\omega_n^{\text{lat}} = 2.0$ rad/s. The trajectory is logged to `ego_trajectory_heading.csv` and appears in green.
- *Vision-based PI* (`detection&pid.py`). The reference is reconstructed at every tick from the RGB camera by the CLRNNet lane detector [38] followed by the inverse-perspective-mapping projection of Section 3.7. The same PI controller of the previous architecture consumes the heading error computed in the body frame. At intersections the LKA hands off to the CARLA Traffic Manager autopilot, as described in Section 3.7. The trajectory is logged to `ego_trajectory_vision.csv` and appears in red.

The longitudinal channel is identical in the three experiments: a discrete PI regulator drives the speed to 30 km/h with the gains obtained from the longitudinal step experiment of Section 3.6. All three trajectories are post-processed by `plot_trajectory.py`, which overlays the recorded reference path and computes two scalar performance metrics:

- The *root-mean-square cross-track error*, $\text{RMSE}_{e_y} = \sqrt{(\frac{1}{N}) \sum_{k=1}^N e_{y[k]}^2}$, which summarises the average tracking accuracy over the run;
- The *maximum absolute cross-track error*, $\max |e_y| = \max_k |e_{y[k]}|$, which captures the worst-case excursion from the centre of the lane.

For the vision-based run, the cross-track error reported by the plotting script is computed *a posteriori* against the same recorded reference path used by the other two runs, even though the controller itself never has access to that path. This makes the three runs comparable on a common metric and isolates the effect of the perception layer from the rest of the control loop. It is also worth stressing that the cross-track error used here as a comparison metric is *not* the error signal of the heading-error and vision-based controllers: those two controllers close their loop on an angle, and the cross-track distance is only computed *off-line* for evaluation purposes.



Figure 4.1: Top-down view of Town06 with the trajectory of the heading-error PI controller drawn in cyan by `draw_trajectory_in_carla.py`. The green square marks the start of the run, the red square marks the end. The vehicle enters from the top of the figure, executes a 90° left turn at the intersection, and proceeds east along the arterial. The line stays well inside the lane throughout the manoeuvre, which is the basic visual confirmation that the controller fulfils its LKA purpose. The remaining discussion of this chapter focuses on the quantitative differences between the three architectures along this

same route.

4.2 Qualitative View in CARLA

Before turning to the numerical metrics, it is useful to look at the test route as it appears in the simulator itself. The script `draw_trajectory_in_carla.py` takes a logged trajectory and draws it as a coloured line directly in the CARLA world, on top of the textured road, using the simulator’s debug-drawing API. The start of the trajectory is marked with a green sphere and the end with a red sphere. Figure 4.1 reproduces one of these renderings for the heading-error PI run, which is the controller that achieves the best tracking accuracy over the full route.

Two observations can already be made from the in-simulator view. First, the trajectory stays well inside the lane throughout the manoeuvre—the rendered line never leaves the asphalt and remains close to the centre of the lane on both the curved and the straight sections of the route. This is the minimum requirement for a Lane Keeping Assist function and the most basic safety check. Second, the corner itself is the most demanding part of the route: it concentrates a large change in the reference heading into a short stretch of road and produces the largest transients in the cross-track error, as the next section will show.

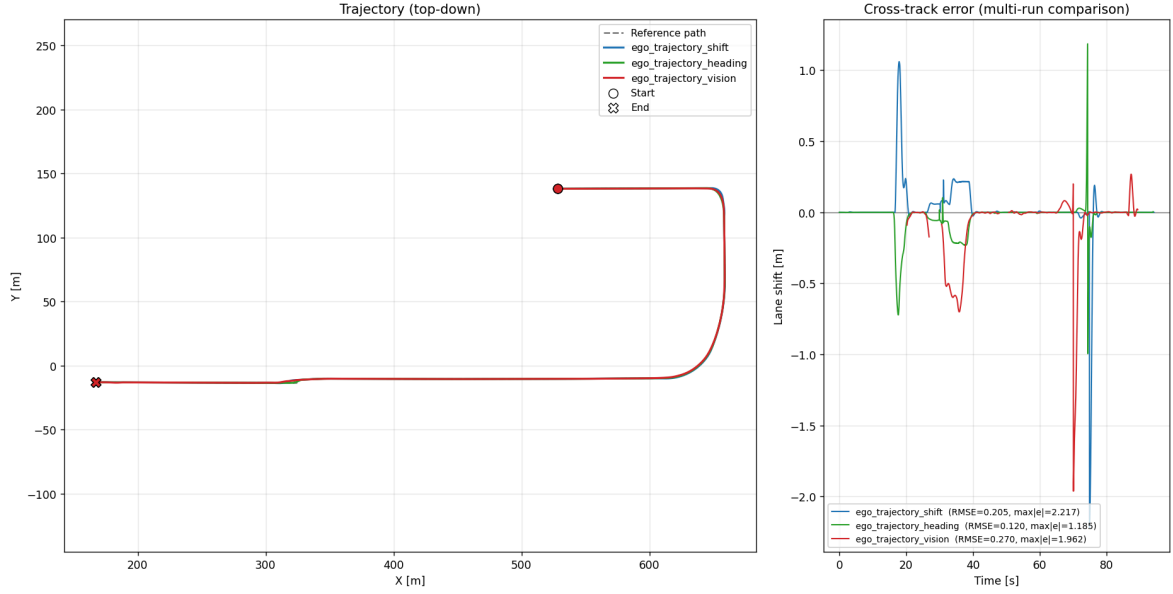


Figure 4.2: Full-route comparison of the three controllers. *Left*: top-down view of the recorded reference path (grey dashed) and of the three closed-loop trajectories (cross-track PID in blue, heading-error PI in green, vision-based PI in red). Start and end of each run are marked with circles and crosses respectively. *Right*: cross-track error $e_y(t)$ as a function of time, with the root-mean-square error and the maximum absolute error reported in the legend. The two large transients around $t \approx 18\text{--}22$ s and $t \approx 70\text{--}75$ s correspond respectively to the 90° corner and to a step-like lane discontinuity in the reference path.

4.3 Full-Route Tracking Performance

The first quantitative comparison is performed over the entire route. Figure 4.2 shows the top-down trajectory plot (left) and the cross-track error as a function of time (right), with the three controllers overlaid on the same axes and with the RMSE and maximum absolute error reported in the legend.

A few comments on the figure. On the long straight section that follows the corner—roughly from $t \approx 25$ s to $t \approx 60$ s—the three trajectories are visually indistinguishable from the reference at the scale of the left-hand panel, even though small but persistent biases are visible on the right-hand error trace. These steady-state biases are the subject of Section 4.4. The two large excursions occur at the 90° corner, where the cross-track and the vision-based controllers overshoot by approximately 1 m, and at the step-like lane discontinuity around $t \approx 70\text{--}75$ s, where the maximum excursions of the run are recorded. Note that the cross-track error trace of the vision-based controller has a finite duration: the controller starts logging only once the lane detector returns a valid centre line, which is the reason why the red curve is missing on the very first seconds of the trace.

The numerical metrics extracted from the right-hand panel are summarised in Table 4.4. The heading-error PI achieves both the lowest RMSE and the lowest maximum excursion of the three architectures, by a clear margin. The cross-track PID is intermediate in RMSE but exhibits the worst maximum excursion of the three runs at the lane discontinuity. The vision-based PI has the highest RMSE, mainly because of a persistent steady-state bias on the smooth curve that the next section will analyse in detail.

Controller	RMSE e_y [m]	$\max e_y $ [m]	Worst-case event
Cross-track PID	0.205	2.217	Step-like lane discontinuity
Heading-error PI	0.120	1.185	Step-like lane discontinuity
Vision-based PI	0.270	1.962	Step-like lane discontinuity

Table 4.4: Tracking metrics over the entire route, read directly from the legend of Figure 4.2. Lower is better.

The relative ordering of the three controllers—heading-error best, cross-track intermediate, vision-based worst on RMSE—matches the qualitative prediction of Table 3.3 on the structural argument (heading-error preferred over cross-track because of the lower plant order) but adds a new piece of information that the theoretical chapter could not anticipate: the perception layer of the vision-based controller introduces a steady-state bias on the smooth curve that dominates its RMSE. The maximum-excursion ordering is different: there, the worst performer is the cross-track PID at 2.22 m, with the vision-based PI at 1.96 m and the heading-error PI at 1.19 m. All three peaks coincide in time with the step-like lane discontinuity of the reference path, which is a deliberately adversarial test case discussed in Section 4.5.

4.4 Performance on a Smooth Curve

The first zoom-in focuses on the long smooth curve that occurs after the corner, between $t \approx 25$ s and $t \approx 45$ s. The road curvature in this section is gentle and approximately constant, so the relevant figure of merit is the *steady-state cross-track error* that each controller maintains while tracking a constant-curvature reference. Figure 4.3 shows the top-down trajectory (left) and the cross-track error (right) restricted to this section of the route.

Three quantitative observations can be made from Figure 4.3. The cross-track PID (blue) settles to a positive steady-state error of approximately +0.22 m, which means that the vehicle rides on the *outside* of the curve relative to the centre line. This is the classical signature of a PID that is fighting a constant disturbance—the road curvature acts as a constant input to the lateral plant—and that has not yet had the time to absorb it through its integral term. The heading-error PI (green) settles to a negative steady-state error of approximately −0.22 m, which means that the vehicle rides on the *inside* of the curve.

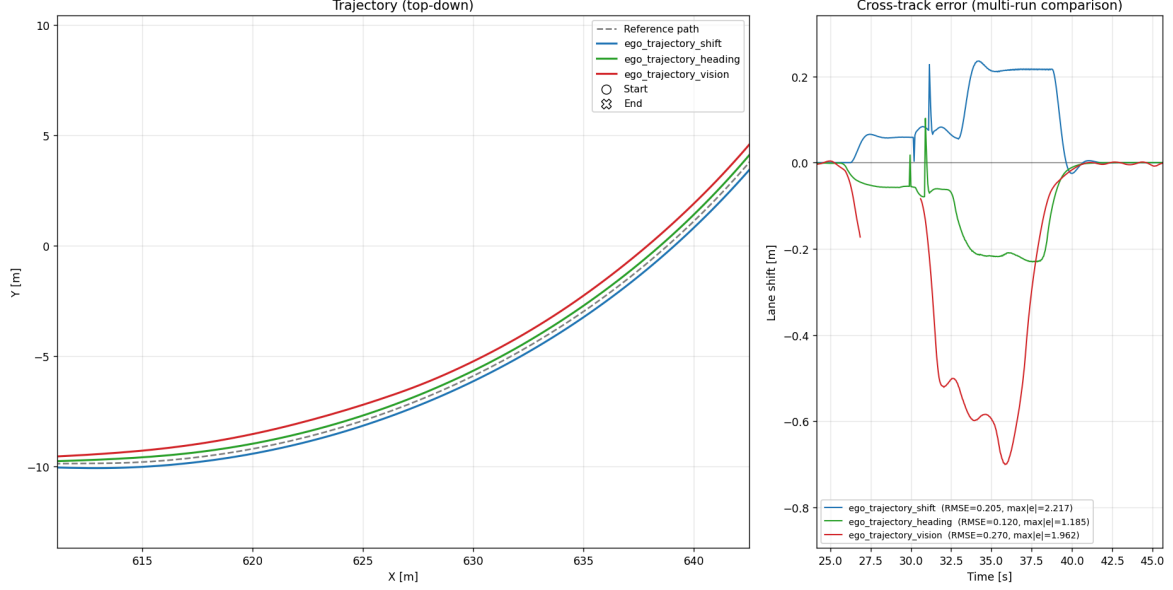


Figure 4.3: Smooth-curve detail. *Left*: top-down view of the recorded reference (grey dashed) and of the three closed-loop trajectories along a gentle, approximately constant-curvature stretch of road between $X \approx 615$ m and $X \approx 640$ m. The three controllers track the same curve with different signed steady-state biases that are clearly visible at this magnification. *Right*: corresponding cross-track error between $t \approx 25$ s and $t \approx 45$ s. The vision-based PI exhibits a large negative bias that reaches ≈ -0.7 m at the apex of the curve, while the cross-track PID and the heading-error PI settle to biases of opposite sign with comparable magnitude of $\approx +0.22$ m and ≈ -0.22 m respectively.

The opposite sign of the two biases is a direct consequence of the different error signal: the cross-track controller tries to keep $e_y = 0$ at the closest waypoint, whereas the heading-error controller tries to keep the look-ahead angle $\alpha = 0$, and the look-ahead point on a curved reference is geometrically inside the curve relative to the closest waypoint. The two biases have similar magnitude but they reflect two different geometric definitions of “being on the centre”.

The vision-based PI (red) exhibits a much larger negative bias that reaches approximately -0.7 m at the apex of the curve, a factor of three larger than the heading-error PI on which it is built. The two controllers share the same control law and the same gains; the only difference is the source of the look-ahead point. The conclusion is that the additional bias is introduced by the perception layer. Two mechanisms contribute to it. First, the centre line returned by `get_centerline` is the average of the two *detected* lane boundaries, which are themselves curved; the mid-point of two parallel arcs is biased toward the inside of the curve by an amount proportional to the curvature and to the lateral spacing between the boundaries. Second, the first-degree polynomial fit of Section 3.7 approximates a locally curved centre line with a straight line, which systematically truncates the lateral component of the look-ahead target. Both mechanisms have the same sign and combine

into the observed bias. On a straight road, neither mechanism is active and the vision-based controller behaves exactly like the heading-error PI, which is consistent with the near-zero error reported in Figure 4.2 before $t \approx 25$ s.

It is also worth noting that the magnitudes involved here— 0.22 m for the path-based controllers and 0.7 m for the vision-based controller—remain comfortably within the half-width of a typical CARLA lane, which is approximately 1.75 m. The biases on the smooth curve therefore do not threaten the safety of the LKA function; they only degrade the comfort and the tracking accuracy.

4.5 Performance on a Sudden Lane Discontinuity

The second zoom-in focuses on a step-like discontinuity of the reference path that occurs around $t \approx 70$ – 75 s, which corresponds to the right-hand spikes visible in Figure 4.2 and which is shown in detail in Figure 4.4. This discontinuity is an artefact of the way the reference path was recorded: when the vehicle in autopilot mode merged from one lane into an adjacent one, the closest-waypoint logger jumped from the centre line of the first lane to the centre line of the second one, which produces a step of approximately one lane width (3 m) in the reference. From the controller’s point of view, this is the worst possible kind of disturbance: a step in the reference rather than a step in the disturbance input.

The three controllers react to the step in three qualitatively different ways. The cross-track PID (blue) follows the original lane up to the discontinuity, then suddenly sees a large positive cross-track error appear on its closest waypoint and reacts with a large steering input that overshoots the new lane by ≈ 2.2 m on the opposite side. This is the worst-case event of the whole run for this controller, and it is the manifestation of the limited phase margin of the cross-track loop: a step on the input of a double integrator is integrated twice before the controller has any chance to react, and the resulting overshoot is bounded only by the rate-limited saturation of the steering actuator. The transient takes approximately 2 s to decay.

The heading-error PI (green) reacts differently. The look-ahead point is, by construction, several metres ahead of the vehicle; the controller therefore “sees” the discontinuity coming and starts steering toward the new lane before the discontinuity is reached. The cross-track error in Figure 4.4 shows a much smaller positive transient of approximately +1 m, followed by a small undershoot of -1 m as the look-ahead point crosses the step in the opposite direction. The transient is over within a single look-ahead time horizon, that is in less than 1 s. The maximum-excursion metric on this controller is the smallest of the three at 1.19 m, which is essentially this single peak.

The vision-based PI (red) reacts earliest of all three because the lane detector reconstructs the upcoming centre line several metres ahead of the vehicle and the polynomial fit smooths the discontinuity into a gradual lateral shift. The cross-track error trace shows

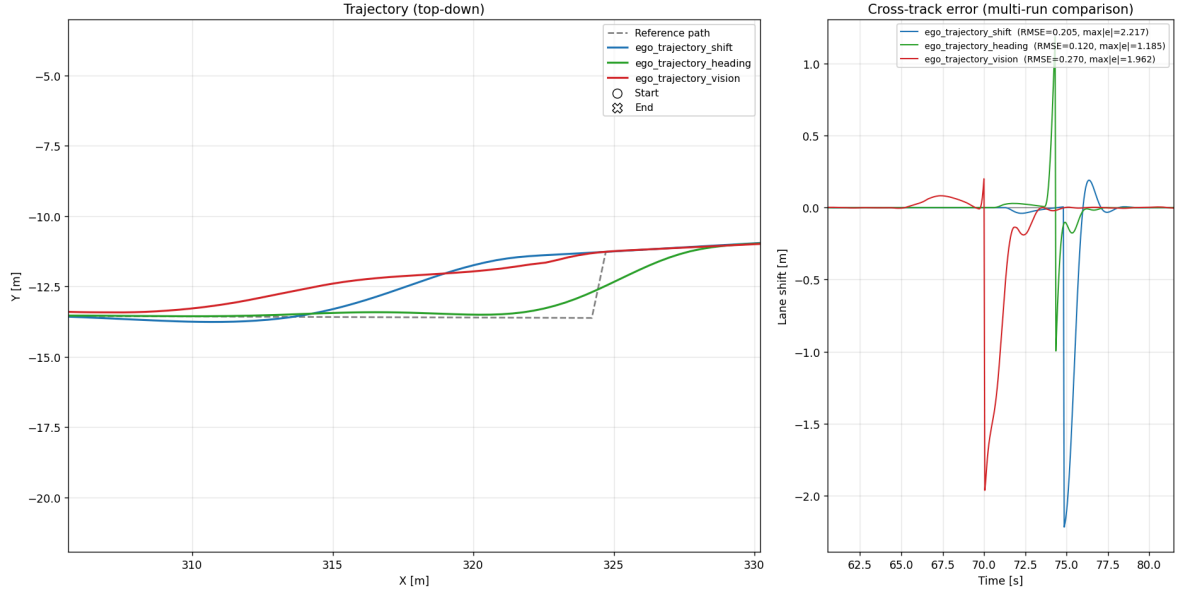


Figure 4.4: Step-like lane discontinuity. *Left*: top-down view of the reference (grey dashed) and of the three closed-loop trajectories in the neighbourhood of the discontinuity. The reference exhibits a step of approximately 3 m at $X \approx 325$ m. The three controllers react with very different transients. *Right*: cross-track error between $t \approx 60$ s and $t \approx 80$ s. The cross-track PID (blue) takes the largest excursion at ≈ -2.22 m; the vision-based PI (red) anticipates the discontinuity and undershoots by ≈ -1.96 m around $t \approx 70$ s; the heading-error PI (green) is the smoothest of the three, with a single overshoot of $\approx +1$ m. a large negative excursion that starts at $t \approx 67$ s—well before the path-based controllers react—and reaches ≈ -2 m at $t \approx 70$ s. This is *not* a tracking failure: the vehicle is correctly following what the camera sees, and what the camera sees is a smooth lateral shift rather than a step. The error is large because the metric is computed against the path-based reference, which itself contains the step. In a real-world deployment, where the only reference is the camera image, this trajectory would actually be the smoothest and the most natural-feeling of the three.

The lane-discontinuity event therefore illustrates the fundamental difference between the three approaches: the path-based controllers track a pre-recorded reference and are penalised by any discontinuity in that reference, while the vision-based controller tracks what is actually painted on the road and can naturally absorb such discontinuities at the cost of an inevitable lag with respect to a discontinuous ground-truth metric.

4.6 Discussion

The numerical results of this chapter both confirm and refine the control-theoretic predictions of Section 3. The reduction of one plant integrator from the cross-track to the heading-error formulation buys a measurable improvement in tracking accuracy: the

heading-error PI achieves an RMSE of 0.120 m against the 0.205 m of the cross-track PID, a factor of 1.7 improvement, and a maximum excursion of 1.19 m against 2.22 m, a factor of 1.9 improvement. Both numbers are consistent with the larger phase margin and the gentler gain schedule that Table 3.3 attributed to the heading-error formulation.

The addition of a perception layer on top of the heading-error PI is less clear-cut. On a straight road, the vision-based controller is indistinguishable from its path-based counterpart, as expected. On a smooth curve, the perception layer introduces a steady-state bias of approximately -0.7 m that the controller cannot remove because it never sees the path-based reference. This bias is the price to pay for being independent of a pre-recorded map. On a step-like lane discontinuity, the perception layer is actually beneficial because it smooths the discontinuity into a gradual lateral shift; the large excursion that appears on the cross-track error metric is an artefact of the comparison against a discontinuous reference, not a tracking failure of the controller.

A subtler point is the role of the controller speed of reaction. The heading-error PI uses a look-ahead distance of $L_d \approx 9$ m at 30 km/h, which gives the controller approximately 1 s of preview on the reference. The vision-based PI uses the same preview horizon but applies it to a perception-based reference. The cross-track PID has no preview at all: it reacts to the closest waypoint, which is by definition the present, not the future. The factor of 1.9 on the maximum excursion between the cross-track and the heading-error controllers is therefore not only a consequence of the lower plant order, but also of the implicit feed-forward that the look-ahead introduces. Disentangling the two effects would require an additional experiment in which the cross-track PID is augmented with an explicit feed-forward term computed from the road curvature; this is left for future work.

From a practical standpoint, the experiments support the following recommendations for an LKA function deployed in this kind of simulation environment. The cross-track PID is the simplest of the three architectures to implement and tune, but it is the most sensitive to discontinuities in the reference and to changes in the vehicle speed; it is acceptable for low-speed, smooth-path applications. The heading-error PI is the most accurate and the most robust to reference discontinuities; it is the recommended choice whenever a pre-recorded reference path is available. The vision-based PI is the only one of the three that does not depend on a pre-recorded map; it pays for this independence with a steady-state bias on curves, which can be mitigated either by a higher-order polynomial fit of the centre line or by a curvature-aware correction of the look-ahead point. The detailed implementation of these mitigations is the subject of the chapter on future work.

The next chapter draws the overall conclusions of the thesis and outlines the directions for future investigation.

Chapter 5

Conclusions and Future Work

This thesis has investigated the implementation and evaluation of a Lane Keeping Assist (LKA) function inside a software-only Vehicle-in-the-Loop simulation environment based on the CARLA simulator. The work spans the full chain from the selection of the simulation platform, through the modelling and the system identification of the vehicle, to the design of three progressively richer lateral-control architectures and to their numerical evaluation on a common test route. The present chapter summarises the main findings of the previous chapters, restates the contributions of the work, and outlines the directions in which the present results could be extended.

5.1 Summary of the Work

The thesis is articulated into four substantive chapters that build on each other in a clear progression from environment selection to controller evaluation.

Section 2 has selected CARLA as the simulation platform of the work, motivated by its open-source nature, its Unreal-Engine-4-based rendering, its accurate physics engine, and—most importantly for the present application—its comprehensive Python API. The same chapter has introduced the dynamic single-track bicycle model that underlies the controller design and has explained how the vehicle parameters m, J, l_f, l_r, c_f, c_r have been read from the CARLA blueprint of the chosen `vehicle.nissan.patrol`. A dispatching procedure that converts the analytical longitudinal acceleration into the throttle/brake commands accepted by the simulator has been described, together with a dual-mode data-gathering pipeline—autopilot for reference paths, manual driving for edge cases—that produces the `.csv` files consumed by the controllers.

Section 3 has developed the lateral-control component of the LKA function in three progressively richer architectures. The cross-track PID closes the loop on the perpendicular distance between the vehicle and a pre-recorded reference path; the corresponding plant is a double integrator and a full PID is required for stability. The heading-error PI closes the loop on the angle between the vehicle’s forward direction and a look-ahead point on the same reference path; the corresponding plant is a single integrator and a PI controller is sufficient. The vision-based PI replaces the pre-recorded reference with a centre line reconstructed at every tick from the RGB camera by the CLRNet lane detector [38] and by an inverse-perspective-mapping projection; the same PI controller of the heading-error formulation consumes the heading error computed in the body frame. A pole-placement

procedure derives the PID gains analytically from a measurement of the steering gain K_{steer} obtained by an open-loop step experiment on the simulator, and predicts that the heading-error formulation should be both more accurate, on a per-bandwidth basis, and more robust to changes in the vehicle speed.

Section 4 has evaluated the three architectures on a common test route in Town06. The numerical results have confirmed the predictions of the theoretical chapter on the relative ordering of the path-based controllers: the heading-error PI achieves an RMSE of 0.120 m and a maximum excursion of 1.185 m against the 0.205 m and 2.217 m of the cross-track PID, an improvement by a factor of approximately 1.7 on the average and of approximately 1.9 on the worst case. The vision-based PI has achieved an RMSE of 0.270 m, which is larger than both path-based controllers; the increase is dominated by a steady-state bias of approximately 0.7 m on the smooth curve, which has been traced back to two specific approximations of the perception layer—the geometric bias of the midline reconstruction and the first-degree polynomial fit of the centre line. On a step-like lane discontinuity, the vision-based controller has been shown to react earlier than the path-based controllers and to produce the smoothest trajectory, at the cost of a large *apparent* error against the discontinuous path-based reference.

5.2 Contributions

The original contributions of this thesis can be summarised as follows.

- A modular, software-only Vehicle-in-the-Loop implementation of a Lane Keeping Assist function in the CARLA simulator, in which the controller runs as an external Python process and communicates with the simulator through the synchronous-mode Python API at 20 Hz. The code is organised into three self-contained modules—`lane_shift_pid/`, `heading_error_pid/`, and the vision-based `detection&pid.py`—that share the same actuator, the same vehicle, and the same simulation rate, so that any observed difference between them can be attributed to the structure of the controller rather than to incidental implementation details. The complete source tree is publicly available in the open-source repository `LaneDet-Carla` [40], which accompanies this thesis.
- A complete, reproducible system-identification procedure (`carla_sysid.py` and `pid_tuning.py`) that fits the parameters of a first-order longitudinal model and a one-parameter lateral steering gain from two open-loop step experiments, and derives analytical PID gains by pole placement on the resulting plants. The procedure is fully automatic and produces ready-to-use Python dictionaries for the controllers.

- A side-by-side comparison of three lateral-control architectures of increasing sophistication, including a detailed control-theoretic analysis (plant order, gain scheduling, phase margin) and a numerical evaluation on a common test route with two scalar metrics. The comparison isolates, for the first time in this specific simulation setup, the contribution of the perception layer from the contribution of the controller structure.
- A vision-based LKA controller that combines a state-of-the-art lane-detection network [38] with an inverse-perspective-mapping projection and with the same heading-error PI of the path-based architecture. The vision-based controller is shown to be a viable LKA implementation that does not require any pre-recorded reference path, at the cost of a steady-state bias on curves that has been quantitatively characterised.
- An explicit junction-handling logic that detects intersections through the CARLA waypoint graph and hands the control over to the Traffic Manager autopilot for the duration of the junction, with a clean reset of the PI controller state at every re-engagement. This makes it possible to run the vision-based LKA on routes that include intersections, which would otherwise be unmanageable for a vision-only formulation.

5.3 Limitations

The work also has limitations that should be acknowledged.

The evaluation is restricted to a single vehicle, a single map (Town06) and a single target speed (30 km/h). The gain-scheduling analysis of Section 3.5 predicts that the heading-error PI is robust to speed variations in the range 20–60 km/h, but this prediction has not been verified experimentally; the same is true of the robustness to vehicle parameters. A systematic sweep over speed, vehicle and map would strengthen the generalisability of the results.

The reference path is recorded once with the CARLA autopilot, which is not guaranteed to be perfectly centred on the lane and which produces a step-like discontinuity at the lane-merge point of the route. This discontinuity has been kept on purpose, because it provides a stress test for the controllers, but a cleaner reference would allow a more direct quantitative comparison.

The perception layer is run in inference-only mode with the pre-trained CLNet weights on the TuSimple dataset. The detector has not been fine-tuned on CARLA images, which produces a residual domain gap that contributes to the steady-state bias on curves. A simple data-collection and fine-tuning step on synthetic CARLA frames would likely reduce this bias.

The longitudinal controller is a simple PI on the speed and has not been the subject of a detailed analysis in this thesis. The cross-track and heading-error chapters have focused on the lateral channel; the coupling between the longitudinal and the lateral dynamics—for instance the effect of braking on the lateral acceleration during a tight turn—has been neglected.

Finally, the entire evaluation is performed in simulation and no real-vehicle test has been carried out. The Sim-to-Real gap is well known to introduce its own set of issues, in particular at the level of the perception layer, and the present results should be considered as indicative of the relative ordering of the three architectures rather than as quantitative predictions of their real-world performance.

5.4 Future Work

Among the many directions in which the present work could be extended, the following three are the ones that the author considers most worth pursuing.

The first is the replacement of the reactive PI controllers with a *Model-Predictive Controller* (MPC). All three architectures presented in this thesis close a feedback loop on a scalar error and react to its present value through a fixed control law; the rate saturation of the steering actuator then sets a hard limit on how aggressively the controllers can recover from a large transient, which is precisely what produces the worst-case excursions reported in Section 4 at the corner and at the lane discontinuity. An MPC that explicitly optimises a finite-horizon cost on the lateral and heading errors, subject to the steering-rate constraint, would shape the transient in a control-theoretically optimal way rather than reacting to it. The bicycle model of Section 3.2 is directly usable as the prediction model, and the system-identification procedure of Section 3.6 already provides the required parameters, so the prediction layer is essentially in place; what is missing is the optimisation back-end and a careful tuning of the cost weights.

The second is the *fusion of the path-based and the vision-based references* into a single look-ahead target. The two extremes have been characterised in detail in this thesis: the path-based controllers are accurate on the recorded route but require a pre-existing map, and the vision-based controller generalises outside the recording but pays a steady-state bias on curves. An intermediate architecture that weighs the two sources by their respective covariances—a Kalman filter on the look-ahead point, or any equivalent multi-sensor fusion scheme—would inherit the accuracy of the path-based controllers where the map is reliable, and would fall back gracefully on the camera elsewhere. The same gains, the same metrics, and the same test route of Section 4 would carry over to the fused architecture, which makes it a natural follow-up rather than a complete rework.

The third, and most ambitious, is the *porting of the framework to a real research vehicle*. The path-based controllers require a localisation module and a pre-recorded

reference; the vision-based controller requires only the camera and the lane detector, and is therefore the most directly portable. A Vehicle-in-the-Loop setup in the strict sense—real vehicle, virtual road generation, synchronised perception—as discussed in the literature review of Section 1, would be the natural intermediate step between the present pure-simulation evaluation and a fully on-road test. The Sim-to-Real gap on the perception layer is the main risk and would itself become a research topic; the control layer, having been derived analytically from a measured plant, should transfer with comparatively few surprises.

These three directions are listed in increasing order of ambition and of required infrastructure. The first two are largely software-only and could be carried out within the existing simulation framework; the third would require a substantial hardware effort and a multi-person team. Together they sketch a research programme that builds incrementally on the contributions of the present thesis.

5.5 Closing Remarks

The work presented in this thesis has shown that a state-of-the-art driving simulator, an external Python control stack, and a deep-learning-based lane detector can be combined into a complete Lane Keeping Assist function that is amenable to rigorous control-theoretic analysis and to quantitative evaluation. The three architectures investigated cover the spectrum from a purely map-based controller to a purely vision-based one, and their side-by-side comparison clarifies the trade-offs that any practical LKA system has to face: simplicity against accuracy, accuracy against generalisation, generalisation against robustness to perception failures. None of the three architectures dominates the others on every metric; each occupies a specific point in the design space and is appropriate for a specific deployment scenario. This kind of structured, simulation-based exploration of the design space is, in the view of the author, one of the most valuable contributions that a Vehicle-in-the-Loop methodology can bring to the development of autonomous driving systems.

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Figure 3.3 Geometric definition of the cross-track error e_y as the signed perpendicular distance between the vehicle position P_v and the line through the closest waypoint P_w in the direction of the lane forward vector $\hat{t}$. Positive values of e_y correspond to the vehicle being on the right of the lane centre with respect to the direction of travel. . . . 29

Figure 3.4 Heading-error / look-ahead geometry. The look-ahead point P_{la} is the closest waypoint on the reference path at distance at least L_d ahead of the vehicle; the heading error α is the signed angle between the vehicle forward unit vector $\hat{f}_v$ and the line $P_v \rightarrow P_{la}$. The look-ahead distance is scheduled as $L_d = L_d^{\min} + k_L v(t)$ 31

Figure 3.5 Block-diagram comparison of the cross-track and heading-error plants. The cross-track formulation contains two integrators—yaw rate to heading, and lateral velocity to cross-track distance—while the heading-error formulation contains only one. The reduction in plant order is the reason why a PI controller is sufficient for the heading-error case, while a full PID is required for the cross-track case. 32

Figure 3.6 Analytical comparison of the cross-track PID and heading-error PI controllers, computed symbolically from the identified plant parameters. *Top*: open-loop magnitude and phase. *Bottom-left*: closed-loop step response. *Bottom-right*: proportional gain as a function of vehicle speed, normalised at 30 km/h. The gentler $\frac{1}{v}$ scaling of the heading-error gain, compared with the $\frac{1}{v^2}$ scaling of the cross-track gain, is the main reason why the heading-error formulation is preferred in the operational range of an LKA function. 35

Figure 3.7 Inverse perspective mapping (side view). A pixel below the horizon back-projects to a unique point on the ground plane, given the camera height h_{cam} and the camera intrinsics (f_x, f_y, c_u, c_v) . Pixels above the horizon $v \leq c_v$ project to infinity and are discarded. 37

Figure 4.1 Top-down view of Town06 with the trajectory of the heading-error PI controller drawn in cyan by `draw_trajectory_in_carla.py`. The green square marks the start of the run, the red square marks the end. The vehicle enters from the top of the figure, executes a 90° left turn at the intersection, and proceeds east along the arterial. The line stays well inside the lane throughout the manoeuvre, which is the basic visual confirmation that the controller fulfils its LKA purpose. The remaining discussion of this chapter focuses on the quantitative differences between the three architectures along this same route. 43

Figure 4.2 Full-route comparison of the three controllers. *Left*: top-down view of the recorded reference path (grey dashed) and of the three closed-loop trajectories (cross-track PID in blue, heading-error PI in green, vision-based PI in red). Start and end of each run are marked with circles and crosses respectively. *Right*: cross-track error $e_{y(t)}$ as a function of time, with the root-mean-square error and the maximum absolute error

reported in the legend. The two large transients around $t \approx 18\text{--}22$ s and $t \approx 70\text{--}75$ s correspond respectively to the 90° corner and to a step-like lane discontinuity in the reference path. 44

Figure 4.3 Smooth-curve detail. *Left*: top-down view of the recorded reference (grey dashed) and of the three closed-loop trajectories along a gentle, approximately constant-curvature stretch of road between $X \approx 615$ m and $X \approx 640$ m. The three controllers track the same curve with different signed steady-state biases that are clearly visible at this magnification. *Right*: corresponding cross-track error between $t \approx 25$ s and $t \approx 45$ s. The vision-based PI exhibits a large negative bias that reaches ≈ -0.7 m at the apex of the curve, while the cross-track PID and the heading-error PI settle to biases of opposite sign with comparable magnitude of $\approx +0.22$ m and ≈ -0.22 m respectively. 45

Figure 4.4 Step-like lane discontinuity. *Left*: top-down view of the reference (grey dashed) and of the three closed-loop trajectories in the neighbourhood of the discontinuity. The reference exhibits a step of approximately 3 m at $X \approx 325$ m. The three controllers react with very different transients. *Right*: cross-track error between $t \approx 60$ s and $t \approx 80$ s. The cross-track PID (blue) takes the largest excursion at ≈ -2.22 m; the vision-based PI (red) anticipates the discontinuity and undershoots by ≈ -1.96 m around $t \approx 70$ s; the heading-error PI (green) is the smoothest of the three, with a single overshoot of $\approx +1$ m. 47

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Appendix A

Carla and Pycharm IDE Interfacing

To control and communicate with Carla, the Python API is used. Therefore, it is necessary to have a Python version compatible with the Carla version we plan to use. However, updates to Carla often require updating the Python version at regular intervals. This makes it mandatory to reconcile Carla and Python with each new update. However, by taking advantage of uv's ability to use different Python versions through different digital environments, we can solve this problem. Thus, we can have multiple environments and Carla models on the same computer, which also reduces the burden of this process. The steps for setting up Carla-uv communication are as follows:

- Downloaded Carla 0.9.16(UE4)
- Installed Pycharm IDE
- Installed uv python package and project manager:
 - 1 `set PROJECT=LaneDetCarla`
 - 2 `powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/0.10.8/install.ps1 | iex"`
 - 3 `git clone https://github.com/axmud/LaneDetCarla.git %PROJECT%`
 - 4 `cd %PROJECT%`
 - 5 `uv venv`
 - 6 `.venv\Scripts\activate`
 - 7 `uv pip install setuptools==66.1.1`
 - 8 `uv pip install wheel==0.46.3`
 - 9 `uv pip install alumentations==0.4.6 --no-build-isolation`
 - 10 `uv sync --extra cu121`
 - 11 `uv pip install --no-build-isolation -e .`
 - 12 `uv pip install numpy==1.23.1`
 - 13 `uv pip install hydra-core==1.3.2`
 - 14 `uv pip install control==0.10.2`

Appendix B

Code Excerpts

This appendix collects the Python excerpts that support Section 2 and Section 3. The full source tree is available in the open-source repository `LaneDetCarla` [40]; what follows is a curated selection of the routines that are referenced explicitly in the text.

BA Data Collectors (Chapter 2)

The two routines used during the data-gathering phase. The first one, `carla_data_collector`, records the dynamic state of the ego vehicle; the second one, `carla_data_collector_2`, records the geometric reference exposed by the CARLA way-point graph. Both are called once per tick from the main loop.

```
1  def carla_data_collector(vehicle: carla.Vehicle):
2      frame = vehicle.get_world().get_snapshot().frame
3      transform = vehicle.get_transform()
4      location = transform.location
5      rotation = transform.rotation
6      velocity = vehicle.get_velocity()
7      acceleration = vehicle.get_acceleration()
8      angular_velocity = vehicle.get_angular_velocity()
9      waypoint = vehicle.get_world().get_map().get_waypoint(location)
10
11     return {
12         "frame": frame,
13         "location": {"x": location.x, "y": location.y, "z":
location.z},
14         "rotation": {"pitch": rotation.pitch, "yaw": rotation.yaw,
"roll": rotation.roll},
15         "velocity": {"x": velocity.x, "y": velocity.y, "z":
velocity.z},
16         "acceleration": {"x": acceleration.x, "y": acceleration.y,
"z": acceleration.z},
17         "angular_velocity": {"x": angular_velocity.x,
18                               "y": angular_velocity.y,
19                               "z": angular_velocity.z},
```

```

20         "waypoint": {"x": waypoint.transform.location.x,
21                     "y": waypoint.transform.location.y,
22                     "z": waypoint.transform.location.z},
23     }
24
25
26     def carla_data_collector_2(vehicle: carla.Vehicle):
27         waypoint = vehicle.get_world().get_map().get_waypoint(
28             vehicle.get_transform().location)
29         loc = waypoint.transform.location
30         fwd = waypoint.transform.get_forward_vector()
31         return {"x": loc.x, "y": loc.y,
32               "vector_x": fwd.x, "vector_y": fwd.y}

```

The duplicate-suppression block that wraps `carla_data_collector_2` to avoid writing identical rows to the `.csv` file when the closest waypoint does not change between consecutive ticks:

```

1     data = {"waypoint_x": [], "waypoint_y": [],
2             "vector_x": [], "vector_y": []}
3
4     while not crashed:
5         world.tick()
6         sample = carla_data_collector_2(ego_vehicle)
7         if (not data["waypoint_x"]
8             or sample["x"] != data["waypoint_x"][-1]
9             or sample["y"] != data["waypoint_y"][-1]):
10             data["waypoint_x"].append(sample["x"])
11             data["waypoint_y"].append(sample["y"])
12             data["vector_x"].append(sample["vector_x"])
13             data["vector_y"].append(sample["vector_y"])
14
15     pd.DataFrame(data).to_csv("carla_data.csv", index=True)

```

The corresponding logging routine for the manual-driving mode. The autopilot is disengaged with `vehicle.set_autopilot(False)` and the user inputs from the keyboard are translated by the `ControlObject` class (also reported in this appendix) into a `carla.VehicleControl` object that is applied to the ego vehicle.

```

1     data = {
2         "frame": [], "x": [], "y": [], "yaw": [],
3         "v_x": [], "v_y": [], "a_x": [],
4         "throttle": [], "brake": [], "steer": []
5     }

```

```

6
7  while not crashed:
8      world.tick()
9      snapshot      = world.get_snapshot()
10     transform     = ego_vehicle.get_transform()
11     velocity      = ego_vehicle.get_velocity()
12     acceleration  = ego_vehicle.get_acceleration()
13     control       = ego_vehicle.get_control()
14
15     data["frame"].append(snapshot.frame)
16     data["x"].append(transform.location.x)
17     data["y"].append(transform.location.y)
18     data["yaw"].append(transform.rotation.yaw)
19     data["v_x"].append(velocity.x)
20     data["v_y"].append(velocity.y)
21     data["a_x"].append(acceleration.x)
22     data["throttle"].append(control.throttle)
23     data["brake"].append(control.brake)
24     data["steer"].append(control.steer)
25
26     pd.DataFrame(data).to_csv("manual_drive_log.csv", index=False)

```

BB Cross-Track Error (Chapter 3, Section 3.4)

The vectorised implementation of Equation (3.23). A monotonically increasing global index is used so that the search remains $O(1)$ in the length of the reference path and so that the controller cannot “snap” backwards on a closed circuit.

```

1  index = 0
2  FUTURE_HORIZON = 10
3
4  def lane_shift_calculator(vehicle: carla.Vehicle):
5      global index
6      loc = vehicle.get_transform().location
7      car_x, car_y = loc.x, loc.y
8
9      end = min(index + FUTURE_HORIZON + 1, len(wp_x_arr))
10     dx = wp_x_arr[index:end] - car_x
11     dy = wp_y_arr[index:end] - car_y
12     sq_dist = dx * dx + dy * dy

```

```

13     offset = int(np.argmin(sq_dist))
14     nearest_index = index + offset
15     index = nearest_index
16
17     vx = vec_x_arr[nearest_index]
18     vy = vec_y_arr[nearest_index]
19     lane_shift = dx[offset] * vy - dy[offset] * vx
20
21     nearest_dist = float(np.sqrt(sq_dist[offset]))
22     return lane_shift, nearest_dist

```

BC Heading Error and Look-Ahead Target (Chapter 3, Section 3.5)

The heading error is computed via a two-argument arctangent so that all four quadrants are handled correctly. The look-ahead target is the first waypoint on the recorded path whose distance from the vehicle exceeds $L_{d(v)} = L_d^{\min} + k_L v$.

```

1  LOOKAHEAD_MIN = 4.0    # m
2  LOOKAHEAD_K   = 0.6    # s
3
4  def get_lookahead_target(vehicle: carla.Vehicle):
5      loc = vehicle.get_transform().location
6      car_x, car_y = loc.x, loc.y
7      v = vehicle.get_velocity()
8      speed_mps = math.sqrt(v.x ** 2 + v.y ** 2)
9      Ld = LOOKAHEAD_MIN + LOOKAHEAD_K * speed_mps
10     Ld_sq = Ld * Ld
11
12     for j in range(index, len(wp_x_arr)):
13         ddx = wp_x_arr[j] - car_x
14         ddy = wp_y_arr[j] - car_y
15         if ddx * ddx + ddy * ddy >= Ld_sq:
16             return float(wp_x_arr[j]), float(wp_y_arr[j]), j
17     last = len(wp_x_arr) - 1
18     return float(wp_x_arr[last]), float(wp_y_arr[last]), last
19
20
21 def heading_error(vehicle: carla.Vehicle, target_x: float, target_y:
    float):

```

```

22     tf = vehicle.get_transform()
23     fwd = tf.get_forward_vector()
24     v_vec = np.array([fwd.x, fwd.y, 0.0])
25     w_vec = np.array([target_x - tf.location.x,
26                       target_y - tf.location.y, 0.0])
27     norm = np.linalg.norm(v_vec) * np.linalg.norm(w_vec)
28     if norm < 1e-6:
29         return 0.0
30     angle = math.acos(np.clip(np.dot(v_vec, w_vec) / norm, -1.0,
31                               1.0))
32     cross_z = v_vec[0] * w_vec[1] - v_vec[1] * w_vec[0]
33     if cross_z < 0:
34         angle = -angle
35     return angle

```

BD PID Controllers (Chapter 3, Section 3.3)

The discrete PID law of Equation (3.21) is implemented as two stateful controllers, one for the lateral channel and one for the longitudinal channel. The state is held in a fixed-length deque, which provides an implicit anti-windup limit equal to the deque length times the sampling period. The output is then clipped to the actuator range.

```

1  class PIDLateralController:
2      def __init__(self, K_P=1.95, K_I=0.05, K_D=0.2, dt=0.05):
3          self._K_P, self._K_I, self._K_D, self._dt = K_P, K_I, K_D,
4          dt
5          self._e_buffer = deque(maxlen=10)
6
7      def run_step(self, heading_err):
8          self._e_buffer.append(heading_err)
9          if len(self._e_buffer) >= 2:
10             de = (self._e_buffer[-1] - self._e_buffer[-2]) /
11             self._dt
12             ie = sum(self._e_buffer) * self._dt
13             else:
14                 de = 0.0
15                 ie = 0.0
16             return float(np.clip(
17                 self._K_P * heading_err + self._K_D * de + self._K_I *
18                 ie,

```

```

16         -1.0, 1.0))
17
18
19 class PIDLongitudinalController:
20     def __init__(self, K_P=0.3, K_I=0.05, K_D=0.0, dt=0.05):
21         self._K_P, self._K_I, self._K_D, self._dt = K_P, K_I, K_D,
dt
22         self._e_buffer = deque(maxlen=10)
23
24     def run_step(self, target_speed, current_speed):
25         error = target_speed - current_speed
26         self._e_buffer.append(error)
27         if len(self._e_buffer) >= 2:
28             de = (self._e_buffer[-1] - self._e_buffer[-2]) /
self._dt
29             ie = sum(self._e_buffer) * self._dt
30         else:
31             de = 0.0
32             ie = 0.0
33         return float(np.clip(
34             self._K_P * error + self._K_D * de + self._K_I * ie,
35             -1.0, 1.0))

```

The combined controller `VehiclePIDController` couples the two channels and returns a single `carla.VehicleControl` object per tick. The signed acceleration produced by the longitudinal controller is mapped to the throttle when positive and to the brake when negative; the steering output of the lateral controller is rate-limited to a maximum increment of 0.1 per step before being applied.

```

1 class VehiclePIDController:
2     def __init__(self, vehicle, args_lateral=None,
args_longitudinal=None,
3         max_throttle=0.75, max_brake=0.3,
max_steering=0.8):
4         self._vehicle = vehicle
5         self._max_throt = max_throttle
6         self._max_brake = max_brake
7         self._max_steer = max_steering
8         self._lat = PIDLateralController(**args_lateral)
9         self._lon = PIDLongitudinalController(**args_longitudinal)
10        self.past_steering = 0.0
11

```

```

12     def run_step_h(self, target_speed, heading_err):
13         v = self._vehicle.get_velocity()
14         current_speed = 3.6 * math.sqrt(v.x**2 + v.y**2 + v.z**2)
15         accel = self._lon.run_step(target_speed, current_speed)
16         steer = self._lat.run_step(heading_err)
17         return self._build_control(accel, steer)
18
19     def _build_control(self, acceleration, steering):
20         control = carla.VehicleControl()
21         if acceleration >= 0.0:
22             control.throttle = min(acceleration, self._max_throt)
23             control.brake = 0.0
24         else:
25             control.throttle = 0.0
26             control.brake = min(abs(acceleration), self._max_brake)
27         if steering > self.past_steering + 0.1:
28             steering = self.past_steering + 0.1
29         elif steering < self.past_steering - 0.1:
30             steering = self.past_steering - 0.1
31         steering = max(-self._max_steer, min(self._max_steer, steering))
32         control.steer = steering
33         self.past_steering = steering
34         return control

```

BE Inverse Perspective Mapping (Chapter 3, Section 3.7)

The `pixel_to_vehicle` routine projects a list of image-plane points onto the ground plane in the body frame of the vehicle, under the pinhole-camera and flat-ground assumptions described in Section 3.7.

```

1  def pixel_to_vehicle(pixel_pts, cam_height,
2                        image_width=1280, image_height=720,
3                        fov_deg=90.0, cam_x_offset=0.0,
4                        max_range=40.0):
5      p = np.asarray(pixel_pts, dtype=np.float64)
6      u, v = p[:, 0], p[:, 1]

```

```

7      fx = fy = image_width / (2.0 * np.tan(np.deg2rad(fov_deg) /
166      2.0))
8      cx, cy = image_width / 2.0, image_height / 2.0
9
10     below_horizon = v > cy + 1e-3
11     u, v = u[below_horizon], v[below_horizon]
12     if u.size == 0:
13         return np.empty((0, 2))
14
15     X_cam = cam_height * fy / (v - cy)
16     Y_cam = X_cam * (u - cx) / fx
17
18     pts = np.column_stack([X_cam + cam_x_offset, Y_cam])
19     return pts[pts[:, 0] <= max_range]

```

The polynomial fit and look-ahead-target evaluation that follow the IPM step:

```

1  def vision_target(centerline_body, Ld, x_min=0.5, x_max=20.0,
166  deg=1):
2      if centerline_body is None or centerline_body.shape[0] < 3:
3          return None
4      pts = centerline_body[(centerline_body[:, 0] > x_min) &
166      (centerline_body[:, 0] < x_max)]
5      if pts.shape[0] < 3:
6          return None
7      coeffs = np.polyfit(pts[:, 0], pts[:, 1], deg=deg)
8      y_at_Ld = float(np.polyval(coeffs, Ld))
9      return Ld, y_at_Ld

```

BF System Identification (Chapter 3, Section 3.6)

The two step experiments and the corresponding nonlinear least-squares fits, as implemented in `carla_sysid.py` and `pid_tuning.py`. The longitudinal model is a first-order step response; the lateral identification uses the steady-state value of the yaw rate to back out the steering gain K_{steer} .

```

1  def fit_longitudinal(lon_df, u_step):
2      t = lon_df['t'].to_numpy()
3      v = lon_df['speed_kmh'].to_numpy()
4
5      def model(t, K, tau):
6          return K * u_step * (1.0 - np.exp(-t / tau))

```

```

7
8     K0 = v[-1] / u_step if u_step > 0 else 60.0
9     idx63 = np.argmax(v >= 0.63 * v[-1])
10    tau0 = max(t[idx63], 0.5)
11    popt, _ = curve_fit(model, t, v, p0=[K0, tau0], maxfev=5000)
12    return float(popt[0]), float(popt[1])
13
14
15    def fit_lateral_gain(lat_df, L, delta_cmd):
16        n = len(lat_df)
17        tail = lat_df.iloc[int(0.75 * n):]
18        psi_dot_ss = math.radians(tail['yaw_rate_dps'].mean())
19        V = tail['speed_kmh'].mean() / 3.6
20        K_steer = psi_dot_ss * L / (V * delta_cmd)
21        return float(K_steer), float(V), float(psi_dot_ss)

```

The pole-placement routines that map the identified plant parameters into the analytical PID gains derived in Equation (3.26) and Equation (3.30):

```

1    def tune_lateral_xtrack(K_lat, wn):
2        # double integrator -> (s+wn)^3 -> full PID
3        K_d = 3.0 * wn / K_lat
4        K_p = 3.0 * wn * wn / K_lat
5        K_i = wn * wn * wn / K_lat
6        return {'K_P': K_p, 'K_I': K_i, 'K_D': K_d}
7
8
9    def tune_lateral_heading(K_lat, wn, zeta=1.0):
10        # single integrator -> (s+wn)^2 -> PI sufficient
11        K_p = 2.0 * zeta * wn / K_lat
12        K_i = wn * wn / K_lat
13        return {'K_P': K_p, 'K_I': K_i, 'K_D': 0.0}
14
15
16    def tune_longitudinal(K, tau, wn):
17        # first-order plant -> (s+wn)^2 -> PI
18        K_p = (2.0 * wn * tau - 1.0) / K
19        K_i = (wn * wn * tau) / K
20        return {'K_P': K_p, 'K_I': K_i, 'K_D': 0.0}

```

BG Manual-Override Class

The `ControlObject` class implements the keyboard fallback used in all three architectures. The class follows a two-step design: the `parse_control` method is called for every keyboard event and only updates the internal flags; the heavier `process_control` method is called once per simulation tick and turns those flags into actual `VehicleControl` values, with progressive throttle ramping, automatic reverse engagement at low speed, and exponential return-to-centre on the steering wheel.

```
1  class ControlObject(object):
2      def __init__(self, veh):
3          self._vehicle = veh
4          self._throttle = False
5          self._brake = False
6          self._steer = None
7          self._steer_cache = 0
8          self._control = carla.VehicleControl()
9
10     def parse_control(self, event):
11         if event.type == pygame.KEYDOWN:
12             if event.key == pygame.K_UP:      self._throttle = True
13             if event.key == pygame.K_DOWN:    self._brake = True
14             if event.key == pygame.K_RIGHT:   self._steer = 1
15             if event.key == pygame.K_LEFT:    self._steer = -1
16         if event.type == pygame.KEYUP:
17             if event.key == pygame.K_UP:      self._throttle = False
18             if event.key == pygame.K_DOWN:
19                 self._brake = False
20                 self._control.reverse = False
21             if event.key in (pygame.K_LEFT, pygame.K_RIGHT):
22                 self._steer = None
23
24     def process_control(self):
25         # Throttle / brake combinations
26         if self._throttle:
27             self._control.throttle = min(self._control.throttle +
0.01, 1)
28             self._control.gear = 1
29             self._control.brake = False
30         elif not self._brake:
31             self._control.throttle = 0.0
```

```

32         if self._brake:
33             if (self._vehicle.get_velocity().length() < 0.01
34                 and not self._control.reverse):
35                 self._control.brake = 0.0
36                 self._control.gear = 1
37                 self._control.reverse = True
38                 self._control.throttle = min(self._control.throttle
+ 0.1, 1)
39             elif self._control.reverse:
40                 self._control.throttle = min(self._control.throttle
+ 0.1, 1)
41             else:
42                 self._control.throttle = 0.0
43                 self._control.brake = min(self._control.brake + 0.3,
1)
44         else:
45             self._control.brake = 0.0
46
47         # Steering with low-pass return-to-centre
48         if self._steer is not None:
49             self._steer_cache += 0.03 * self._steer
50             self._steer_cache = max(-0.7, min(0.7,
self._steer_cache))
51             self._control.steer = round(self._steer_cache, 1)
52         else:
53             self._steer_cache *= 0.2
54             if abs(self._steer_cache) < 0.01:
55                 self._steer_cache = 0.0
56             self._control.steer = round(self._steer_cache, 1)
57
58         self._vehicle.apply_control(self._control)

```